ARC-SPRUCE: Post-Quantum Blind Issuance and Anonymous Rate-Limited Credentials from Lattices

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ABSTRACT

Blind issuance protocols and anonymous credentials are useful for privacy-preserving online authentication systems. ARC-SPRUCE gives a post-quantum issuer-view-hiding blind-issuance construction and a conditional anonymous rate-limiting credential construction from lattice assumptions and hash-based zero knowledge. The construction combines a BB-tran-style proof-friendly signature layer, an Ajtai commitment, a Gaussian preimage-sampling interface, SmallWood NIZKs, and a Poseidon2-style PRF assumption.

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1 INTRODUCTION

Anonymous credentials (AC) [1–3] allow parties to convey certified information about themselves in a privacy-preserving way. The usual roles are issuers, holders, and verifiers. A holder obtains attributes certified by an issuer and later proves statements about those attributes to a verifier without revealing the credential itself. The central security goals are unforgeability and unlinkability: verifiers should only accept credentials derived from an issuer, while issuers and verifiers should not be able to link a presentation to an issuance session or to another presentation.

Certain AC use cases, such as anonymous rate limiting [8, 9], require that a credential be usable only a bounded number of times in a given context, for example per epoch or per service. Existing classical constructions can provide compact credentials and presentations, but practically efficient post-quantum ACs remain much less developed. This gap is relevant for digital identity systems, including wallet deployments that will eventually require post-quantum migration.

Building ACs. Anonymous credentials are often built from blind-signature-style issuance. The holder first commits to the message to be certified, the issuer signs a target derived from this committed message, and the holder later presents a zero-knowledge proof of knowledge of a valid message-signature pair. A generic post-quantum combination of standard signatures, commitments, and zero-knowledge proofs is currently too large for the settings targeted here. We therefore combine a proof-friendly lattice signature, an Ajtai-style commitment, and SmallWood zero-knowledge proofs in a relation that is designed to be compact.

1.1 Contributions

We construct a post-quantum blind-issuance and anonymous rate-limited credential design from lattice assumptions and hash-based zero knowledge. The construction uses proof-friendly lattice signatures from vanishing SIS with rational hashing [4], an NTRU-style Gaussian preimage-sampling candidate based on Antrag [5], and the SmallWood proof system for relatively small statements [6]. The DKLW import is used only for the non-blind hash-and-preimage signature layer. The interactive issuance protocol is treated separately as a correct and issuer-view-hiding blind-issuance layer, with one-more unforgeability left as an explicit assumption in the ARC soundness decomposition. Rate limiting is implemented by signing a hidden PRF key and proving correct tag derivation during presentation.

The blind-issuance layer uses an Ajtai commitment and a two-party randomness step. The holder commits to the full signed message and its holder-side randomness before the issuer sends its contribution. The final rational-hash randomness is obtained by centering the sum of the two contributions. This preserves the shared- randomness issuance architecture, while the rational hash itself is the BB-tran form of DKLW:

$$T = B_0 + B_1\mu + B_2 r_0 + Z, \qquad (B_3 - \mathbf{1}_n r_1) \odot Z = 1.$$

The target is computed from the bare hidden witness elements, not from the commitment value. The commitment serves to bind the holder to its message and holder-side randomness before the issuer randomness is known.

The privacy of the public target is obtained by choosing the linear-side randomizer r_0 long enough for the factorization-aware cyclotomic leftover-hash bound of Boudgoust–Lapiha to apply. Under the stated splitting parameters of R_q, this makes $B_2 r_0$ statistically close to uniform and therefore masks the message-dependent and inverse-side offsets. The inverse-side term is used only as an additive offset in this argument; no bounded-noise claim is made for $(B_3 - \mathbf{1}_n r_1)^{-1}$. The current draft proves correctness and honest-issuer public-surface hiding for the issuance layer and then gives a conditional ARC security decomposition from that issuance layer, NIZK knowledge soundness/zero knowledge, and the concrete PRF assumption.

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1.2 Technical overview

The full signed message is

$$\mu = \mathbf{m} \| \mathbf{k},$$

where $\mathbf{m}$ is the attribute vector and $\mathbf{k}$ is the secret PRF key.

Issuance. The holder samples $\mathbf{k}$, holder-side hash randomness r_0^H, r_1^H, and commitment-only randomness $\bar{r}$. It sends the Ajtai commitment

$$\mathbf{c} = \mathbf{A}_{\mathsf{Com}} [\mu \| r_0^H \| r_1^H \| \bar{r}]^\top$$

to the issuer. The issuer samples bounded randomness r_0^I, r_1^I and returns it. The holder computes

$$r_0 = \mathsf{center}(r_0^H + r_0^I), \qquad r_1 = \mathsf{center}(r_1^H + r_1^I),$$

then derives

$$Z = (B_3 - \mathbf{1}_n r_1)^{-1}, \qquad T = B_0 + B_1 \mu + B_2 r_0 + Z.$$

The holder sends T together with a pre-sign NIZK proving that the same bounded witness opens the commitment, computes the centered randomness, and satisfies the BB-tran target relation. If the proof verifies, the issuer samples a short preimage $\mathbf{u} \leftarrow \mathsf{SampPre}(\mathsf{td}, T)$ such that $\mathbf{Au} = T$ and returns it to the holder.

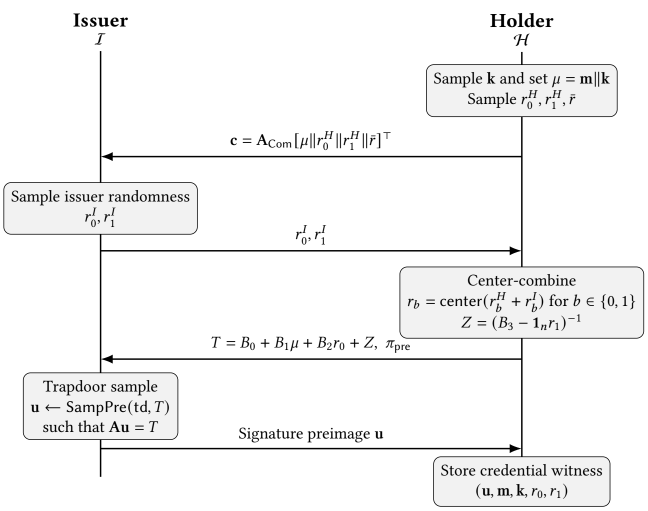

Figure 1: Issuance flow. The pre-sign proof binds the BB-tran target to the committed message and the centered shared randomness, while hiding the holder witness.

Showing. To present a credential, the holder chooses a nonce, computes $\mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce})$, and proves knowledge of $\mathbf{u}, \mathbf{m}, \mathbf{k}, r_0, r_1, Z$ such that

$$\begin{aligned} (B_3 - \mathbf{1}_n r_1) \odot Z &= 1, \\ \mathbf{Au} &= B_0 + B_1(\mathbf{m}\|\mathbf{k}) + B_2 r_0 + Z, \\ \mathsf{tag} &= \mathsf{F}(\mathbf{k}, \mathsf{nonce}). \end{aligned}$$

The verifier sees only $(\mathsf{nonce}, \mathsf{tag}, \pi_{\mathsf{Show}})$ and enforces the rate policy by recording accepted $(\mathsf{nonce}, \mathsf{tag})$ pairs.

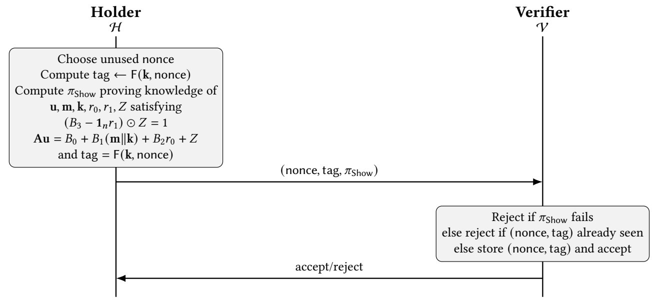

Figure 2: Showing flow. The tag is deterministic for a fixed key and nonce, enabling stateful rate limiting, while the proof hides the credential witness.

SmallWood proofs. We use SmallWood [6] as the NIZK for the issuance and showing relations. Both relations are expressed over committed witness rows packed on an evaluation domain. The issuance proof checks the Ajtai opening, the centering equations, and the BB-tran target relation. The showing proof adds the short preimage equation and the PRF-tag relation. Section 5 states the semantic compiled relations and the coefficient-domain consequences used by the security arguments; Appendix C records the schematic row inventories and proof-stack details.

2 PRELIMINARIES

This section fixes notation and recalls the formal syntax of the lattice primitives, the BB-tran rational hash, non-interactive zero-knowledge arguments, blind signatures, anonymous rate-limited credentials, and pseudorandom functions. We also record the imported non-blind signature-security statements from Dubois–Klooß–Lai–Woo (DKLW) that are used later for the hash-and-preimage signature layer.

2.1 Notation

Let λ denote the security parameter and let $\mathrm{negl}(\lambda)$ be any positive function negligible in λ. We write PPT for probabilistic polynomial time. For integers $a < b$, let $[a, b] = \{a, a+1, \ldots, b\}$ and use $[b]$ as shorthand for $[1, b]$. Vectors are written in bold lower-case letters and matrices in bold upper-case letters.

We work over the cyclotomic ring

$$R = \mathbb{Z}[X]/(X^N + 1)$$

for power-of-two N, and its quotient

$$R_q = R/qR.$$

We identify R_q with polynomials of degree $< N$ with coefficients in $\mathbb{Z}_q$. For $a \in \mathbb{Z}$, we write $\langle a \rangle_q \in \{0, \ldots, q-1\}$ for reduction modulo q. Whenever a field element is interpreted as a signed integer, we use its centered representative in $[-q/2, q/2]$. For $x \in R_q$, the notation $\|x\|_\infty \leq B$ means that every coefficient of x lies in $[-B, B]$; the same convention is used component-wise for vectors over R_q.

The full signed message is written

$$\mu = \mathbf{m} \| \mathbf{k},$$

where $\mathbf{m}$ encodes holder attributes and $\mathbf{k}$ is the secret PRF key used for rate limiting. The key is sampled from a finite key space $\mathcal{K}_{\text{key}} \subseteq \mathbb{F}_q^{\ell_{\text{key}}}$ fixed by the public parameters. In the bounded-key instantiation used by the present relations, we require

$$\mathcal{K}_{\text{key}} \subseteq \{\mathbf{k} \in \mathbb{F}_q^{\ell_{\text{key}}} : \|\mathbf{k}\|_\infty \leq \beta_{\text{msg}}\},$$

and the PRF assumption is made for a uniform key from this same key space. The public coefficient bound for holder attributes and protocol randomness is denoted β_{msg}, and the short-preimage bound for signatures is denoted $\beta_{\texttt{sig}}$.

Whenever the CRT decomposition $R_q \cong \prod_i F_i$ is used, we write

$$(x_i)_i \odot (y_i)_i := (x_i y_i)_i, \qquad (x_i)_i \oslash (y_i)_i := (x_i y_i^{-1})_i$$

whenever every $y_i \neq 0$. Thus $\odot$ and $\oslash$ denote slotwise multiplication and slotwise division in CRT coordinates.

2.2 Centering map and shared randomness

The issuance protocol combines holder-side and issuer-side bounded randomness by a coefficient-wise centering operation.

Definition 2.1 (Centering map). Fix the coefficient bound β_{msg} and let

$$M_{\text{ctr}} := 2\beta_{\text{msg}} + 1.$$

For an integer $a \in \mathbb{Z}$, define

$$\mathsf{center}(a) \in [-\beta_{\text{msg}}, \beta_{\text{msg}}]$$

to be the unique representative of a modulo M_{ctr} in that interval. Extend center coefficient-wise from integers to ring elements of R_q and then component-wise to vectors over R_q.

Lemma 2.2 (Uniformity of centered sums). *Let $x^H \in [-\beta_{\text{msg}}, \beta_{\text{msg}}]$ be fixed and let x^I be uniform over $[-\beta_{\text{msg}}, \beta_{\text{msg}}]$. Then*

$$x = \mathsf{center}(x^H + x^I)$$

is uniform over $[-\beta_{\text{msg}}, \beta_{\text{msg}}]$. The same holds coefficient-wise for ring elements and vectors over R_q.

Proof. The map $x^I \mapsto \mathsf{center}(x^H + x^I)$ is a permutation of the interval $[-\beta_{\text{msg}}, \beta_{\text{msg}}]$ because it is translation modulo M_{ctr}. Therefore it preserves the uniform distribution. Applying this argument coefficient-wise proves the ring and vector statements. □

2.3 Commitment schemes and Ajtai commitments

Definition 2.3 (Commitment scheme). A non-interactive commitment scheme is a tuple of PPT algorithms

$$\mathsf{Com} = (\mathsf{Setup}, \mathsf{Commit}, \mathsf{Verify})$$

with the following syntax.

- $\mathsf{pp}_{\mathsf{Com}} \leftarrow \mathsf{Setup}(1^\lambda, \mathsf{pp})$ outputs public commitment parameters.
- $(c, d) \leftarrow \mathsf{Commit}(m)$ takes a message m and outputs a commitment c and an opening d.
- $\mathsf{Verify}(c, m, d) \in \{0, 1\}$ deterministically checks whether d is a valid opening of c to m.

The scheme is correct if honestly generated commitments always verify. It is computationally binding if no PPT adversary can open one commitment to two distinct messages except with negligible probability. It is hiding if commitments to any two messages are computationally indistinguishable.

2.3.1 *Ajtai commitment used in ARC-SPRUCE.*

Fix message length ℓ_μ, holder-randomness lengths ℓ_{r_0}, ℓ_{r_1}, commitment-only randomness length $\ell_{\bar{r}}$, and output length ℓ_{Com}. Let

$$\ell_{\text{in}} := \ell_\mu + \ell_{r_0} + \ell_{r_1} + \ell_{\bar{r}}.$$

The public commitment matrix is

$$\mathbf{A}_{\text{Com}} \xleftarrow{\$} R_q^{\ell_{\text{Com}} \times \ell_{\text{in}}}.$$

When useful, we block-parse it as

$$\mathbf{A}_{\text{Com}} = [\mathbf{A}_\mu \mid \mathbf{A}_{r_0} \mid \mathbf{A}_{r_1} \mid \mathbf{A}_{\bar{r}}],$$

so that the message part and the three randomness blocks are explicit.

A holder committing to the full message μ and to holder-side rational-hash randomness (r_0^H, r_1^H) samples additional commitment-only randomness $\bar{r}$ and computes

$$\mathbf{c} = \mathbf{A}_{\text{Com}}[\mu \| r_0^H \| r_1^H \| \bar{r}]^\top. \tag{1}$$

Equivalently,

$$\mathbf{c} = \mathbf{A}_\mu \mu + \mathbf{A}_{r_0} r_0^H + \mathbf{A}_{r_1} r_1^H + \mathbf{A}_{\bar{r}} \bar{r}.$$

The opening is $(\mu, r_0^H, r_1^H, \bar{r})$. Verification checks the linear equation (1) together with the coefficient bounds on all opened values.

Definition 2.4 (Module-SIS and Module-ISIS over R_q). Fix integers $n, m \geq 1$, modulus $q \geq 2$, and bound $\beta > 0$.

- $\mathsf{M\text{-}SIS}_\infty(n, m, q, \beta)$. Given $\mathbf{A} \in R_q^{n \times m}$, find a non-zero $\mathbf{s} \in R^m$ such that $\mathbf{A}\mathbf{s} = 0$ in R_q^n and $\|\mathbf{s}\|_\infty \leq \beta$.
- $\mathsf{M\text{-}ISIS}_\infty(n, m, q, \beta)$. Given $(\mathbf{A}, \mathbf{t}) \in R_q^{n \times m} \times R_q^n$, find $\mathbf{s} \in R^m$ such that $\mathbf{A}\mathbf{s} = \mathbf{t}$ in R_q^n and $\|\mathbf{s}\|_\infty \leq \beta$.

Lemma 2.5 (Binding of the Ajtai commitment). *If a PPT adversary opens one Ajtai commitment to two distinct bounded vectors with non-negligible probability, then there exists a PPT adversary that solves the corresponding* $\mathsf{M\text{-}SIS}_\infty$ *instance with essentially the same probability and runtime.*

Proof. Suppose two distinct valid openings

$$(\mu, r_0^H, r_1^H, \bar{r}) \neq (\mu', r_0^{H\prime}, r_1^{H\prime}, \bar{r}')$$

verify the same commitment $\mathbf{c}$. Subtracting the two opening equations gives

$$\mathbf{A}_{\text{Com}}\left([\mu - \mu' \| r_0^H - r_0^{H\prime} \| r_1^H - r_1^{H\prime} \| \bar{r} - \bar{r}']^\top\right) = 0.$$

The difference vector is non-zero and coefficient-wise bounded by $2\beta_{\text{msg}}$, hence it is an $\mathsf{M\text{-}SIS}_\infty$ solution. □

Lemma 2.6 (Statistical hiding of the Ajtai commitment). *Assume q is unramified in R and write the factorization*

$$X^N + 1 \equiv \prod_{j=1}^{f_{\text{fac}}} \Phi_j(X) \pmod{q}, \qquad \deg \Phi_j = d_{\text{fac}}, \qquad N = f_{\text{fac}} d_{\text{fac}}.$$

Let the commitment-only randomness $\bar{r}$ be sampled coefficient-wise from $[-B_r, B_r]$ with $B_r < q/2$. Define

$$\varepsilon_{\mathsf{Com}} := \frac{1}{2}\sqrt{\left(\left(\frac{q^{\ell_{\mathsf{Com}}}}{(2B_r+1)^{\ell_{\bar{r}}}}\right)^{d_{\mathrm{fac}}}+1\right)^{f_{\mathrm{fac}}} - 1}.$$

Then the commitment distribution is statistically $\varepsilon_{\mathsf{Com}}$-close to a distribution independent of the committed vector (μ, r_0^H, r_1^H). The commitment parameters are λ-hiding when $\varepsilon_{\mathsf{Com}} \le 2^{-\lambda}$. In the bounded instantiation used here, one may take $B_r = \beta_{\mathtt{msg}}$.

Proof. For fixed (μ, r_0^H, r_1^H), the commitment is a public offset plus the linear image $\mathbf{A}_{\bar{r}}\bar{r}$. Applying the bounded-uniform cyclotomic leftover-hash lemma with input dimension $\ell_{\bar{r}}$, output dimension ℓ_{Com}, and coefficient radius B_r gives the displayed statistical-distance bound. Adding the fixed offset does not change statistical distance. □

2.4 Lattice trapdoors and rational hashing

We recall the interface used by the hash-and-preimage signature layer. Additional lattice-assumption syntax, such as vSIS, hinted-vSIS, and GenISIS$_f$, is deferred to the appendix.

Definition 2.7 (Admissible lattice trapdoor suite)**.** A lattice trapdoor suite over R_q consists of PPT algorithms

$$(\mathsf{Setup}, \mathsf{TrapGen}, \mathsf{SampPre})$$

together with a public bound $\beta_{\mathtt{sig}}$. We say that the suite is admissible for the parameters used in this paper if the following interface holds.

- $\mathsf{pp}_{\mathsf{td}} \leftarrow \mathsf{Setup}(1^\lambda, \mathsf{pp})$ outputs dimensions (n, m), modulus q, sampler parameters, and the accepted shortness bound $\beta_{\mathtt{sig}}$.
- $(\mathbf{A}, \mathsf{td}) \leftarrow \mathsf{TrapGen}(\mathsf{pp}_{\mathsf{td}})$ outputs a public matrix $\mathbf{A} \in R_q^{n\times m}$ together with a trapdoor td, and the distribution of $\mathbf{A}$ is statistically or computationally close to uniform over $R_q^{n\times m}$ as required by the invoked signature theorem.
- On input $(\mathsf{pp}_{\mathsf{td}}, \mathsf{td}, T)$ with $T \in R_q^n$, the sampler $\mathsf{SampPre}(\mathsf{td}, T)$ outputs $\mathbf{u} \in R^m$ such that $\mathbf{Au} = T$ in R_q^n, except with negligible failure probability, and $\|\mathbf{u}\|_\infty \le \beta_{\mathtt{sig}}$ except with negligible probability.
- The distributional hiding property required by the DKLW hash-and-preimage signature theorem holds for the produced preimages at the chosen Gaussian width and public bound.

This definition is an interface assumption for the signature-layer import. Appendix B describes the NTRU/Antrag-based candidate used for parameter tuning, but Antrag alone is not cited here as a proof of this full admissible-suite interface.

Definition 2.8 (Trapdoor rational hash)**.** A trapdoor rational hash scheme over R_q is a tuple

$$\mathsf{TDRH} = (\mathsf{Setup}, \mathsf{TrapGen}, \mathsf{Eval}, \mathsf{SampPre})$$

consisting of the lattice trapdoor suite above together with a public evaluation algorithm. We extend Setup so that $\mathsf{pp}_{\mathsf{td}}$ also contains a public hash description B, a randomness space $\mathcal{X}$, a message space $\mathcal{M}$, and an admissibility error bound δ. For fixed B, the algorithm

$$\mathsf{Eval}(\mathsf{pp}_{\mathsf{td}}, \mu, \chi)$$

takes a message $\mu \in \mathcal{M}$ and witness randomness $\chi \in \mathcal{X}$ and outputs either a target $T \in R_q^n$ or $\perp$ if the input is inadmissible. For a public description B contained in $\mathsf{pp}_{\mathsf{td}}$, define the admissible message subspace

$$\mathcal{M}_B := \left\{\mu \in \mathcal{M} : \Pr_{\chi\leftarrow\mathcal{X}}\left[\mathsf{Eval}(\mathsf{pp}_{\mathsf{td}}, \mu, \chi) = \perp\right] \le \delta\right\}.$$

Correctness requires that for every $\mu \in \mathcal{M}_B$ and every $\chi \in \mathcal{X}$ such that $\mathsf{Eval}(\mathsf{pp}_{\mathsf{td}}, \mu, \chi) = T \neq \perp$, the output $\mathbf{u} \leftarrow \mathsf{SampPre}(\mathsf{td}, T)$ satisfies

$$\mathbf{Au} = T \qquad \text{and} \qquad \|\mathbf{u}\|_\infty \le \beta_{\mathtt{sig}}$$

except with negligible probability.

The generic trapdoor-rational-hash interface does not by itself imply that the public target hides the signed message. In the concrete BB-tran instantiation used in this paper, target hiding is obtained later from a cyclotomic leftover-hash lemma applied to the long linear-side randomizer.

The BB-tran instantiation. We instantiate the rational hash with the BB-tran family from DKLW [4, Table 1]. In that work the generic expression is

$$h_{\mu,\chi}(\tilde{b}^\top) = (u^\top, x_0^\top)\tilde{b}_0 + \frac{1}{\tilde{b}_1 - x_1}.$$

The translation used in ARC-SPRUCE is

$$u \leftrightarrow \mu, \qquad x_0 \leftrightarrow r_0, \qquad x_1 \leftrightarrow r_1.$$

To account for the affine term used in the protocol, we view the DKLW linear side as having an implicit leading coordinate:

$$(1, \mu^\top, r_0^\top)\widehat{B}_0 = B_0 + B_1\mu + B_2 r_0.$$

Thus the public description is

$$B = (B_0, B_1, B_2, B_3),$$

where $B_0, B_3 \in R_q^n$ and B_1, B_2 are public linear maps into R_q^n. The inverse-side randomizer $r_1 \in R_q$ is broadcast to all target coordinates; for readability we write $B_3 - r_1$ for $B_3 - \mathbf{1}_n r_1$. If an implementation instead uses a vector-valued $r_1 \in R_q^n$, that is a vectorized variant and its DKLW predicate and security side conditions must be assumed separately.

For message μ and randomness (r_0, r_1), define the linear side

$$L_B(\mu, r_0) := B_0 + B_1\mu + B_2 r_0. \tag{2}$$

The input (B, μ, r_0, r_1) is admissible if $B_3 - \mathbf{1}_n r_1 \in R_q^{\times n}$, equivalently if every CRT slot of every coordinate of $B_3 - \mathbf{1}_n r_1$ is non-zero. When admissible, let

$$Z = (B_3 - \mathbf{1}_n r_1)^{-1}$$

with inversion understood coordinate-wise in CRT form. The BB-tran target is then

$$T = h_{\mathrm{tran},\mu,(r_0,r_1)}(B) := L_B(\mu, r_0) + Z. \tag{3}$$

Equivalently, the witness form is

$$(B_3 - \mathbf{1}_n r_1) \odot Z = 1, \qquad T = B_0 + B_1\mu + B_2 r_0 + Z, \tag{4}$$

and the eliminated form is

$$(B_3 - \mathbf{1}_n r_1) \odot \left(T - B_0 - B_1\mu - B_2 r_0\right) = 1. \tag{5}$$

The witness form is preferred in the protocol statements because it isolates the bilinear inverse check from the linear identity for the public target.

Lemma 2.9 (Eliminated form under admissibility). *Let $R_q \cong \prod_i F_i$ be the CRT decomposition and write $B_3 - \mathbf{1}_n r_1 = (d_i)_i$ and $T - L_B(\mu, r_0) = (z_i)_i$. If $d_i \neq 0$ for all CRT slots, then*

$$T = h_{\mathrm{tran},\mu,(r_0,r_1)}(B) \iff (B_3 - \mathbf{1}_n r_1) \odot \big(T - B_0 - B_1\mu - B_2 r_0\big) = 1.$$

Proof. In each CRT slot, the definition is the scalar statement $z_i = d_i^{-1}$, which is equivalent to $d_i z_i = 1$ when $d_i \neq 0$. Reassembling the slots proves the equivalence. ◻

Definition 2.10 (Long target-randomizer). Fix an integer $\ell_{r_0} \geq 1$. The linear-side BB-tran randomizer is sampled as

$$r_0 \xleftarrow{\$} [-\beta_{\mathsf{msg}}, \beta_{\mathsf{msg}}]^{N\ell_{r_0}} \subseteq R_q^{\ell_{r_0}},$$

meaning that each coefficient of each ring coordinate of r_0 is sampled independently and uniformly from $[-\beta_{\mathsf{msg}}, \beta_{\mathsf{msg}}]$.

Lemma 2.11 (Cyclotomic leftover-hash specialization). *Assume q is unramified in R and write*

$$X^N + 1 \equiv \prod_{j=1}^{f_{\mathrm{fac}}} \Phi_j(X) \pmod q, \qquad \deg \Phi_j = d_{\mathrm{fac}}, \qquad N = f_{\mathrm{fac}} d_{\mathrm{fac}}.$$

Let $n \geq 1$, let $B_2 \xleftarrow{\$} R_q^{n \times \ell_{r_0}}$, and let r_0 be distributed as in Definition 2.10. Define

$$\varepsilon_{r_0} := \frac{1}{2}\sqrt{\left(\left(\frac{q^n}{(2\beta_{\mathsf{msg}}+1)^{\ell_{r_0}}}\right)^{d_{\mathrm{fac}}} + 1\right)^{f_{\mathrm{fac}}} - 1}.$$

Then

$$\Delta\big((B_2, B_2 r_0), (B_2, U)\big) \leq \varepsilon_{r_0}, \qquad U \xleftarrow{\$} R_q^n.$$

In particular, the parameters give λ-bit target hiding from this step when $\varepsilon_{r_0} \leq 2^{-\lambda}$.

Proof. Apply the bounded-uniform cyclotomic leftover-hash lemma to the linear map $r_0 \mapsto B_2 r_0$ with input dimension ℓ_{r_0}, output dimension n, and coefficient radius β_{msg}. The displayed expression is the resulting statistical-distance bound for the stated factorization of R_q. ◻

Theorem 2.12 (LHL-based target hiding of h_{tran}). *Assume the hypotheses of Lemma 2.11, so the public parameters are honestly generated and in particular B_2 is sampled uniformly at setup. Let r_1 be any fixed admissible inverse-side randomizer and define, for a bounded message $\mu \in \mathcal{M}$,*

$$T_\mu := B_0 + B_1\mu + B_2 r_0 + (B_3 - \mathbf{1}_n r_1)^{-1}.$$

Then T_μ is statistically ε_{r_0}-close to uniform over R_q^n. Consequently, for any two bounded messages $\mu_0, \mu_1 \in \mathcal{M}$,

$$\Delta(T_{\mu_0}, T_{\mu_1}) \leq 2\varepsilon_{r_0}.$$

If $\varepsilon_{r_0} \leq 2^{-\lambda}$, this gives $\Delta(T_{\mu_0}, T_{\mu_1}) \leq 2^{1-\lambda}$.

Proof. By Lemma 2.11, the distribution of $B_2 r_0$ is statistically ε_{r_0}-close to a uniform $U \xleftarrow{\$} R_q^n$. For fixed admissible r_1 and fixed message μ, the remaining term

$$C_{\mu,r_1} := B_0 + B_1\mu + (B_3 - \mathbf{1}_n r_1)^{-1}$$

is a deterministic offset, hence

$$T_\mu = B_2 r_0 + C_{\mu,r_1}$$

is also statistically ε_{r_0}-close to uniform. Applying the same argument to μ_0 and μ_1 and using the triangle inequality yields the displayed bound. No coefficient-norm bound on $(B_3 - \mathbf{1}_n r_1)^{-1}$ is used in this hiding argument: the inverse-side term enters only as an additive offset. ◻

Corollary 2.13 (Protocol distribution of the target randomizer). *Assume $r_0^H, r_0^I \in R_q^{\ell_{r_0}}$ are sampled coefficient-wise in $[-\beta_{\mathsf{msg}}, \beta_{\mathsf{msg}}]$, with r_0^I uniform. Then*

$$r_0 = \mathsf{center}(r_0^H + r_0^I)$$

is coefficient-wise uniform over $[-\beta_{\mathsf{msg}}, \beta_{\mathsf{msg}}]$ by Lemma 2.2. Hence the protocol distribution of r_0 satisfies Definition 2.10. If issuance restarts until $B_3 - \mathbf{1}_n r_1$ is admissible, the target-hiding argument remains valid after conditioning on admissibility because Theorem 2.12 is pointwise in every fixed admissible r_1.

2.5 Signatures and imported non-blind security

The blind-issuance layer in this paper builds on the non-blind vSIS hash-and-preimage signature layer from DKLW. We therefore record the standard signature syntax and the strong unforgeability notions used by that import.

Definition 2.14 (Signature scheme). A signature scheme for a message space $\mathcal{M}$ is a tuple of PPT algorithms

$$\Sigma = (\mathsf{Setup}, \mathsf{KG}, \mathsf{Sign}, \mathsf{Verify}).$$

It is correct if honestly generated signatures verify except with negligible probability.

Definition 2.15 (Strong existential unforgeability). Let Σ be a signature scheme. We use the standard notions of strong existential unforgeability under chosen-message attack (sEUF-CMA), selective-message attack (sEUF-SMA), and strong selective unforgeability under selective-message attack (sSUF-SMA), exactly as in the DKLW formalization.

Definition 2.16 (vSIS-based hash-and-preimage signature). Let TDRH be a trapdoor rational hash scheme with parameters $(B, \mathcal{M}, \mathcal{X}, \delta, \beta_{\mathtt{sig}})$. The associated vSIS signature scheme

$$\mathsf{vSIS} = (\mathsf{KG}, \mathsf{Sign}, \mathsf{Verify})$$

is defined as follows.

- KG runs $(\mathbf{A}, \mathsf{td}) \leftarrow \mathsf{TrapGen}(\mathsf{pp}_{\mathsf{td}})$ and outputs $(\mathsf{vk}, \mathsf{sk}) = (\mathbf{A}, \mathsf{td})$.
- $\mathsf{Sign}(\mathsf{sk}, \mu)$ samples $\chi \leftarrow \mathcal{X}$ until $T = \mathsf{Eval}(\mathsf{pp}_{\mathsf{td}}, \mu, \chi) \neq \perp$, then samples $\mathbf{u} \leftarrow \mathsf{SampPre}(\mathsf{td}, T)$ and outputs
$$\sigma = (\chi, \mathbf{u}).$$
- $\mathsf{Verify}(\mathsf{vk}, \mu, \sigma)$ for $\sigma = (\chi, \mathbf{u})$ checks that $\chi \in \mathcal{X}$, $\mu \in \mathcal{M}_B$, $\|\mathbf{u}\|_\infty \leq \beta_{\mathtt{sig}}$, and
$$\mathbf{A}\mathbf{u} = \mathsf{Eval}(\mathsf{pp}_{\mathsf{td}}, \mu, \chi).$$

Theorem 2.17 (Specialized DKLW security import). *Let Σ_{tran} be the vSIS signature scheme obtained from Definition 2.16 by instantiating the rational family with the BB-tran form of* (3). *Assume the admissible trapdoor-suite interface of Definition 2.7, the DKLW predicate and denominator-bound conditions, the required strong-linear-independence condition for the instantiated rational functions, the required norm-compatibility conversion between DKLW's norm* *convention and the public bound $\beta_{\mathtt{sig}}$, negligible inadmissibility errors, and the corresponding strong hinted-vSIS hypothesis. Then Σ_{tran} is strongly existentially unforgeable under selective-message attack.*

Proof. This is the BB-tran specialization of the DKLW selective-query security theorem for their generic vSIS-based signature family. The statement here is conditional on the DKLW side conditions being satisfied by the concrete ARC-SPRUCE parameters. ◻

Corollary 2.18 (Conditional imported route to sEUF-CMA)**.** *Suppose, in addition, that the exact hypotheses of the DKLW $\mathsf{GenISIS}_f/\mathsf{IntGenISIS}_f$ lifting theorem for BB-tran are met, including the required power-of-two cyclotomic setting, the modulus and message-bound condition $q/2 > \gamma$, the dimension condition $m = n \log q + \omega(\log \lambda)$, the smoothing lower bound on the Gaussian parameter, and the theorem's stated bound loss. Then the same non-blind signature layer Σ_{tran} obtains the corresponding strong existential unforgeability guarantee under adaptive chosen-message attack.*

Proof. This is a conditional import of DKLW's GenISIS/IntGenISIS lifting theorem. It concerns only the non-blind hash-and-preimage signature layer and does not imply one-more unforgeability of the interactive issuance protocol. ◻

Remark 2.19 (Scope of the import). Theorems 2.17 and 2.18 concern the non-blind hash-and-preimage signature layer only. They do not by themselves prove one-more unforgeability of the blind issuance protocol.

2.6 Non-interactive zero-knowledge arguments of knowledge

We use the NIZK-ARK syntax of the SmallWood framework instantiated in the random-oracle model.

Definition 2.20 (NIZKs in the random-oracle model)**.** Let $\mathcal{R} \subseteq \mathcal{X} \times \mathcal{W}$ be a family of relations. A transparent non-interactive zero-knowledge argument of knowledge for $\mathcal{R}$ in the random-oracle model, with oracle RO, is a pair of algorithms $(\mathsf{Prove}^{\mathsf{RO}}, \mathsf{Verify}^{\mathsf{RO}})$:

- $\mathsf{Prove}^{\mathsf{RO}}(w \mid \mathcal{R}, x) \to \pi$ is probabilistic and, on input a statement-witness pair $(x, w) \in \mathcal{R}$, outputs a proof π.
- $\mathsf{Verify}^{\mathsf{RO}}(\pi \mid \mathcal{R}, x) \to \{0, 1\}$ is deterministic.

It satisfies the following properties.

Completeness. For every $(x, w) \in \mathcal{R}$,

$$\Pr\left[\mathsf{Verify}^{\mathsf{RO}}(\pi \mid \mathcal{R}, x) = 1 : \pi \leftarrow \mathsf{Prove}^{\mathsf{RO}}(w \mid \mathcal{R}, x)\right] = 1.$$

Straightline-extractable knowledge soundness. There exists a PPT extractor Ext such that for every PPT adversary $\mathcal{A}$,

$$\Pr\left[\mathsf{Verify}^{\mathsf{RO}}(\pi \mid \mathcal{R}, x) = 1 \ \wedge \ (x, w) \notin \mathcal{R}\right] \leq \varepsilon,$$

where $(\pi, Q) \leftarrow \mathcal{A}^{\mathsf{RO}}(x)$, Q is the set of random-oracle query/answer pairs made by $\mathcal{A}$, and $w \leftarrow \mathsf{Ext}(Q, \pi)$.

Zero knowledge. There exists a PPT simulator Sim, which may program the random oracle, such that for every $(x, w) \in \mathcal{R}$ and every PPT distinguisher $\mathcal{A}$,

$$\left|\Pr[\mathcal{A}^{\mathsf{RO}}(\pi_{\mathrm{real}}) = 1] - \Pr[\mathcal{A}^{\mathsf{RO}}(\pi_{\mathrm{sim}}) = 1]\right| \leq \xi,$$

where

$$\pi_{\mathrm{real}} \leftarrow \mathsf{Prove}^{\mathsf{RO}}(w \mid \mathcal{R}, x), \qquad \pi_{\mathrm{sim}} \leftarrow \mathsf{Sim}^{\mathsf{RO}}(x).$$

We instantiate these proofs through the SmallWood PACS/PIOP framework compiled with Fiat–Shamir in the random-oracle model.

2.7 Blind signatures

We follow the standard blind-signature syntax.

Definition 2.21 (Blind-signature syntax)**.** A blind signature scheme is a tuple

$$\mathsf{BSig} = (\mathsf{Setup}, \mathsf{IssuerKeyGen}, \mathsf{Issue}, \mathsf{Verify})$$

where Setup generates public parameters, IssuerKeyGen generates issuer keys $(\mathsf{vk}, \mathsf{sk})$, Issue is an interactive protocol between issuer and holder, and $\mathsf{Verify}(\mathsf{vk}, \mu, \mathsf{sig})$ checks signatures.

Definition 2.22 (Blind-signature correctness)**.** A blind signature scheme is correct if there exists negl such that for every message μ,

$$\Pr\left[\begin{array}{l} \mathsf{pp}_{\mathsf{BSig}} \leftarrow \mathsf{Setup}(1^\lambda); \\ (\mathsf{vk}, \mathsf{sk}) \leftarrow \mathsf{IssuerKeyGen}(\mathsf{pp}_{\mathsf{BSig}}); \\ \mathsf{sig} \leftarrow \mathcal{H}^{\mathsf{Issue}:\langle \mathcal{I}(\mathsf{sk}):\cdot \rangle}(\mu, \mathsf{vk}); \\ \mathsf{Verify}(\mathsf{vk}, \mu, \mathsf{sig}) = 1 \end{array}\right] \geq 1 - \mathrm{negl}(\lambda).$$

Definition 2.23 (Blindness)**.** A blind signature scheme is blind if, for any PPT issuer-side adversary interacting with two honest holders on challenge messages μ_0, μ_1, the completed signatures are computationally indistinguishable up to their output order.

Definition 2.24 (One-more unforgeability)**.** A blind signature scheme is one-more unforgeable if any PPT adversary participating in at most Q issuance interactions with the honest issuer cannot output $Q + 1$ valid signatures on distinct messages, except with negligible probability.

Remark 2.25 (Scope of the issuance layer). The protocol layer studied in Section 3 is used as an issuer-view-hiding blind-issuance protocol together with a raw post-issuance verification relation. The current draft does not claim that its completed output instantiates Definition 2.23, and one-more unforgeability of that interactive issuance layer remains a separate reduction goal.

2.8 Anonymous Rate-Limiting Credentials

Rate limiting is part of the ARC syntax: the holder presents a credential together with a public nonce and a PRF-derived public tag, while the verifier enforces the policy by maintaining state on previously accepted $(\mathsf{nonce}, \mathsf{tag})$ pairs.

Definition 2.26 (ARC syntax)**.** Let $\mathcal{N}$ be a finite nonce set. An anonymous rate-limiting credential scheme is a tuple

$$\mathsf{ARC} = (\mathsf{Setup}, \mathsf{IssuerKeyGen}, \mathsf{Issue}, \mathsf{Show}, \mathsf{Verify})$$

where Issue outputs a credential, $\mathsf{Show}(\mathsf{cred}, \mathsf{nonce})$ outputs a presentation, and $\mathsf{Verify}(\mathsf{vk}, \mathsf{pres}, \mathsf{state})$ checks the presentation and updates verifier state.

Definition 2.27 (ARC soundness)**.** Let ARC have nonce space $\mathcal{N}$. After at most Q issuance interactions, no PPT adversary should be able to produce more than $Q|\mathcal{N}|$ accepted presentations, except with negligible probability.

Definition 2.28 (Presentation unlinkability)**.** An ARC scheme has presentation unlinkability if, for distinct fresh nonces, presentations produced from the same credential are computationally indistinguishable from presentations produced from independently issued credentials, up to the intended linkage caused by nonce reuse.

Remark 2.29 (Scope of the ARC theorems). Section 4 does not prove Definition 2.27 in full. It records a conditional policy-soundness statement and a conditional presentation-unlinkability statement for the ARC construction induced by the current issuance layer.

2.9 Pseudorandom functions

Definition 2.30 (Pseudorandom function)**.** Let $\mathcal{K}, \mathcal{D}, \mathcal{T}$ be finite sets. A family

$$\mathsf{F} : \mathcal{K} \times \mathcal{D} \to \mathcal{T}$$

is pseudorandom if oracle access to $\mathsf{F}(k, \cdot)$ for uniform $k \xleftarrow{\$} \mathcal{K}$ is computationally indistinguishable from oracle access to a uniform random function from $\mathcal{D}$ to $\mathcal{T}$.

We instantiate the PRF with a Poseidon2-style permutation, but the PRF security of the keyed feed-forward/truncated family below is an explicit assumption rather than a theorem derived from the Poseidon2 permutation paper.

Definition 2.31 (Poseidon2-style permutation)**.** Let q be prime, let $d \geq 3$ satisfy $\gcd(d, q-1) = 1$, and let $R_F, R_P \in \mathbb{N}$ with R_F even. A Poseidon2-style permutation alternates full external rounds and partial internal rounds, together with public round constants and invertible linear mixing maps over $\mathbb{F}_q$.

Definition 2.32 (Poseidon2-style PRF)**.** For key $\mathbf{k} \in \mathcal{K}_{\mathsf{key}} \subseteq \mathbb{F}_q^{\ell_{\mathsf{key}}}$ and public input $\mathbf{n} \in \mathbb{F}_q^{\ell_{\mathsf{nonce}}}$, define

$$\mathsf{F}(\mathbf{k}, \mathbf{n}) = \mathrm{Tr}_{\ell_{\mathrm{tag}}}\big(P(\mathbf{k}\|\mathbf{n}) + (\mathbf{k}\|\mathbf{n})\big),$$

where P is the Poseidon2-style permutation and $\mathrm{Tr}_{\ell_{\mathrm{tag}}}$ returns the first ℓ_{tag} lanes.

Assumption 2.33 (Concrete Poseidon2-style PRF assumption)**.** *For the concrete parameter choices fixed in Appendix E, the family*

$$\mathsf{F}(k, n) = \mathrm{Tr}_{\ell_{\mathrm{tag}}}\big(P(k\|n) + (k\|n)\big)$$

of Definition 2.32 is a secure pseudorandom function for $k \xleftarrow{\$} \mathcal{K}_{\mathsf{key}}$*. If the signed key space is changed, this assumption is made for the changed key distribution and the ARC relations must enforce that same key space.*

3 BLIND-ISSUANCE CONSTRUCTION

This section presents the blind-issuance protocol that underlies ARC-SPRUCE. The construction combines the Ajtai commitment of Section 2.3, the BB-tran rational hash of Section 2.4, and the SmallWood NIZK-ARK of Section 2.6. The signed message is the full vector $\mu = \mathbf{m}\|\mathbf{k}$.

3.1 Randomizing the BB-tran witness

The holder first commits to

$$(\mu, r_0^H, r_1^H, \bar{r}),$$

where r_0^H and r_1^H are holder-side contributions to the final BB-tran randomness and $\bar{r}$ is commitment-only randomness. The centered value $r_0 \in R_q^{\ell_{r_0}}$ is the long target-randomizer on the linear side of the BB-tran target; its width is chosen so that Theorem 2.12 applies. The issuer then samples bounded randomness

$$r_0^I,\ r_1^I$$

and sends it to the holder. The holder derives the final BB-tran randomness by coefficient-wise centering:

$$r_0 = \mathsf{center}(r_0^H + r_0^I), \qquad r_1 = \mathsf{center}(r_1^H + r_1^I). \tag{6}$$

Whenever $B_3 - \mathbf{1}_n r_1 \in R_q^{\times n}$, the holder computes the inverse witness

$$Z = (B_3 - \mathbf{1}_n r_1)^{-1}$$

and the public BB-tran target

$$T = B_0 + B_1\mu + B_2 r_0 + Z = h_{\mathrm{tran},\mu,(r_0,r_1)}(B). \tag{7}$$

The pre-sign proof therefore binds the public target to the same hidden message and holder randomness already fixed by the commitment, while the issuer controls only the additive randomness through (r_0^I, r_1^I).

3.2 Protocol relations

We record the two relations used later in the issuance and ARC layers.

Pre-sign issuance relation. The public statement is

$$x_{\mathrm{IssueBS}} = (\mathsf{pp}_{\mathsf{Com}}, \mathbf{c}, r_0^I, r_1^I, T, B),$$

and the hidden witness is

$$w_{\mathrm{IssueBS}} = (\mu, r_0^H, r_1^H, \bar{r}, r_0, r_1, Z).$$

Define

$$\mathcal{R}_{\mathrm{IssueBS}} = \Big\{(x_{\mathrm{IssueBS}}, w_{\mathrm{IssueBS}}) : \text{all of the following hold}\Big\}$$

where the constraints are

$$\mathbf{c} = \mathbf{A}_{\mathsf{Com}}[\mu\|r_0^H\|r_1^H\|\bar{r}]^\top, \tag{8}$$

$$r_0 = \mathsf{center}(r_0^H + r_0^I), \tag{9}$$

$$r_1 = \mathsf{center}(r_1^H + r_1^I), \tag{10}$$

$$(B_3 - \mathbf{1}_n r_1) \odot Z = 1, \tag{11}$$

$$T = B_0 + B_1\mu + B_2 r_0 + Z, \tag{12}$$

and

$$\|\mu\|_\infty,\ \|r_0^H\|_\infty,\ \|r_1^H\|_\infty,\ \|\bar{r}\|_\infty,\ \|r_0\|_\infty,\ \|r_1\|_\infty \leq \beta_{\mathtt{msg}}.$$

Thus the proof certifies that the same hidden message and holder randomness both open the commitment and give rise, after combining with the issuer randomness, to the public BB-tran target.

Post-issuance signature relation. For later knowledge statements, it is convenient to isolate the raw signature relation. The public statement is

$$x_{\mathrm{SigBS}} = (\mu, \mathbf{A}, B, \beta_{\mathtt{msg}}, \beta_{\mathtt{sig}}),$$

and the hidden witness is

$$w_{\mathrm{SigBS}} = (\mathbf{u}, r_0, r_1, Z).$$

Define

$$\mathcal{R}_{\mathrm{SigBS}} = \Big\{(x_{\mathrm{SigBS}}, w_{\mathrm{SigBS}}) : (B_3 - \mathbf{1}_n r_1) \odot Z = 1,\ \mathbf{A}\mathbf{u} = B_0 + B_1\mu + B_2 r_0 + Z,\ \|\mathbf{u}\|_\infty \leq \beta_{\mathtt{si}}$$

Equivalently, this states that

$$\mathbf{A}\mathbf{u} = h_{\mathrm{tran},\mu,(r_0,r_1)}(B)$$

with all hidden values lying in the accepted bounded domains.

3.3 Blind-issuance protocol and raw verification

We now record the shared-randomness blind-issuance protocol together with a raw post-issuance verification algorithm for the extracted witness relation.

$\mathsf{Setup}(1^\lambda)$:

(1) Run

$$\mathsf{pp}_{\mathsf{Com}} \leftarrow \mathsf{Com.Setup}(1^\lambda, \mathsf{pp})$$

to obtain the Ajtai commitment key.

(2) Run

$$\mathsf{pp}_{\mathsf{td}} \leftarrow \mathsf{TDRH.Setup}(1^\lambda, \mathsf{pp})$$

to obtain the BB-tran hash description B, the rational-hash randomness space $\mathcal{X}$, the admissibility error bound δ, and the shortness bound $\beta_{\mathtt{sig}}$.

(3) Output

$$\mathsf{pp}_{\mathsf{BSig}} = (\mathsf{pp}_{\mathsf{Com}}, \mathsf{pp}_{\mathsf{td}}, \beta_{\mathsf{msg}}, \beta_{\mathtt{sig}}).$$

$\mathsf{IssuerKeyGen}(\mathsf{pp}_{\mathsf{BSig}})$:

(1) Compute

$$(\mathbf{A}, \mathsf{td}) \leftarrow \mathsf{TDRH.TrapGen}(\mathsf{pp}_{\mathsf{td}}).$$

(2) Output the issuer verification and signing keys

$$\mathsf{vk} = \mathbf{A}, \qquad \mathsf{sk} = \mathsf{td}.$$

Issue: The issuer $\mathcal{I}$ holds sk while the holder $\mathcal{H}$ has input message μ and issuer key vk.

(1) The holder samples holder-side randomness

$$r_0^H,\ r_1^H,\ \bar{r}$$

coefficient-wise in $[-\beta_{\mathsf{msg}}, \beta_{\mathsf{msg}}]$, computes the commitment

$$\mathbf{c} = \mathbf{A}_{\mathsf{Com}}[\mu \| r_0^H \| r_1^H \| \bar{r}]^\top,$$

and sends $\mathbf{c}$ to the issuer.

(2) The issuer samples issuer-side randomness

$$r_0^I,\ r_1^I$$

coefficient-wise in $[-\beta_{\mathsf{msg}}, \beta_{\mathsf{msg}}]$ and sends (r_0^I, r_1^I) to the holder.

(3) The holder computes the centered values r_0, r_1 as in (6). If $B_3 - \mathbf{1}_n r_1 \notin R_q^{\times n}$, the holder aborts the current issuance and restarts from the issuer-randomness round with fresh issuer randomness; this failure probability is accounted for as part of the rational-hash admissibility error. Otherwise it sets

$$Z = (B_3 - \mathbf{1}_n r_1)^{-1}, \qquad T = B_0 + B_1\mu + B_2 r_0 + Z.$$

(4) The holder computes the pre-sign NIZK

$$\pi_{\mathsf{Issue}} \leftarrow \mathsf{NIZK.Prove}\big(w_{\mathrm{IssueBS}} \mid \mathcal{R}_{\mathrm{IssueBS}}, x_{\mathrm{IssueBS}}\big)$$

and sends $(T, \pi_{\mathsf{Issue}})$ to the issuer.

(5) The issuer verifies π_{Issue} against $\mathcal{R}_{\mathrm{IssueBS}}$. If verification fails, the issuer aborts. Otherwise it samples

$$\mathbf{u} \leftarrow \mathsf{TDRH.SampPre}(\mathsf{td}, T)$$

and sends $\mathbf{u}$ to the holder.

(6) The holder checks

$$\mathbf{A}\mathbf{u} = T \qquad \text{and} \qquad \|\mathbf{u}\|_\infty \le \beta_{\mathtt{sig}}.$$

If either check fails, it aborts. Otherwise it outputs the raw post-issuance witness

$$\mathsf{sig} = (\mathbf{u}, r_0, r_1).$$

$\mathsf{Verify}(\mathsf{vk}, \mu, \mathsf{sig})$: On input

$$\mathsf{sig} = (\mathbf{u}, r_0, r_1),$$

verification rejects if any of

$$\|\mathbf{u}\|_\infty > \beta_{\mathtt{sig}}, \qquad \|\mu\|_\infty > \beta_{\mathsf{msg}}, \qquad \|r_0\|_\infty > \beta_{\mathsf{msg}}, \qquad \|r_1\|_\infty > \beta_{\mathsf{msg}},$$

fails. It also rejects if $B_3 - \mathbf{1}_n r_1 \notin R_q^{\times n}$. Otherwise it computes

$$Z = (B_3 - \mathbf{1}_n r_1)^{-1}$$

and accepts iff

$$\mathbf{A}\mathbf{u} = B_0 + B_1\mu + B_2 r_0 + Z.$$

The output $\mathsf{sig} = (\mathbf{u}, r_0, r_1)$ is a raw post-issuance witness for the BB-tran signature relation. The ARC layer uses it only inside later zero-knowledge showing proofs; the current section does not identify it with an unlinkably rerandomized completed blind signature.

3.4 Correctness and issuer-view hiding

Proposition 3.1 (Correctness). *Assume correctness of the NIZK and correctness of the trapdoor sampler. Then honest execution of* Issue *outputs a raw post-issuance witness accepted by* Verify *except with negligible probability.*

Proof. By construction, the holder computes r_0, r_1, Z, T satisfying

$$(B_3 - \mathbf{1}_n r_1) \odot Z = 1, \qquad T = B_0 + B_1\mu + B_2 r_0 + Z.$$

Completeness of the pre-sign NIZK makes the issuer accept, and correctness of SampPre yields $\mathbf{A}\mathbf{u} = T$ with $\|\mathbf{u}\|_\infty \le \beta_{\mathtt{sig}}$ except with negligible failure probability. The final verification algorithm recomputes the same Z and therefore accepts. □

Proposition 3.2 (Honest-issuer public-surface hiding). *Assume:*

(1) the issuer samples r_0^I, r_1^I honestly and independently from the prescribed bounded uniform distributions;
(2) the pre-sign NIZK is zero knowledge;
(3) the Ajtai commitment is hiding as in Lemma 2.6;
(4) the centering map satisfies Lemma 2.2; and
(5) Theorem 2.12 holds for the protocol distribution of r_0.

Then the public issuance surface

$$(\mathbf{c}, r_0^I, r_1^I, T, \pi_{\mathsf{Issue}})$$

is computationally hiding with respect to the holder message μ, up to the NIZK simulation error, the commitment-hiding error, the target-hiding error, and the stated admissibility/restart error.

Proof. Let μ_0, μ_1 be two challenge messages and let $b \leftarrow \{0,1\}$ be the hidden bit. We consider an honest issuance to μ_b.

First replace the real pre-sign proof π_{Issue} by a simulated proof for the same public statement. By zero knowledge, this changes the issuer's distinguishing advantage by at most the NIZK simulation error. After this step, the proof transcript is no longer a possible leakage channel, so the relevant public view is reduced to

$$(\mathbf{c}, r_0^I, r_1^I, T).$$

Second, the Ajtai commitment $\mathbf{c}$ hides the committed vector

$$(\mu_b, r_0^H, r_1^H, \bar{r})$$

by Lemma 2.6. In particular, it hides both the challenge message and the holder-side randomness that will later be combined with the issuer randomness.

Third, the issuer contributions r_0^I, r_1^I are sampled independently of the challenge bit. For every fixed holder-side contribution, Lemma 2.2 implies that

$$r_0 = \mathsf{center}(r_0^H + r_0^I)$$

has exactly the coefficient-wise bounded uniform distribution required by Lemma 2.11. If the protocol restarts until $B_3 - \mathbf{1}_n r_1$ is admissible, the conditioning depends only on r_1 and does not change the bounded-uniform distribution of the independently sampled r_0.

For honestly generated public parameters, and in particular for uniformly sampled B_2, Theorem 2.12 implies that for every fixed admissible value of r_1 the public target

$$T = B_0 + B_1\mu_b + B_2 r_0 + (B_3 - \mathbf{1}_n r_1)^{-1}$$

statistically hides μ_b. Averaging this pointwise bound over the accepted distribution of r_1 preserves the same bound. Since r_0^I and r_1^I are independent of b, combining commitment hiding, independent issuer randomness, and the LHL-based target-hiding theorem proves the stated public-surface hiding claim. □

Remark 3.3 (Security status). The non-blind signature layer has the conditional DKLW guarantees recorded in Section 2.5. The blind-issuance protocol established in this section provides correctness and honest-issuer public-surface hiding. A separate reduction or assumption is still needed for one-more unforgeability of the interactive blind-issuance layer itself.

4 FROM BLIND ISSUANCE TO ANONYMOUS RATE-LIMITING CREDENTIALS

This section lifts the blind-issuance protocol of Section 3 to an anonymous rate-limiting credential (ARC). The main idea is that the full signed message is

$$\mu = \mathbf{m} \| \mathbf{k},$$

where $\mathbf{m}$ carries the holder attributes and $\mathbf{k}$ is a secret PRF key. Issuance signs this full message, while showing proves both signature validity and correct tag derivation

$$\mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce})$$

for a public nonce nonce.

4.1 Why the signed key gives rate limiting

For a fixed credential witness, the hidden key $\mathbf{k}$ is fixed once issuance completes and is required to lie in the same key space $\mathcal{K}_{\mathsf{key}}$ used in Assumption 2.33. Consequently, for any public nonce nonce, the public tag

$$\mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce})$$

is uniquely determined. If the holder reuses the same nonce with the same credential witness, the same tag must reappear. For distinct fresh nonces, Assumption 2.33 says that the resulting tags are computationally indistinguishable from outputs of a random function at those inputs. The verifier can therefore enforce a rate policy by maintaining local state on previously accepted (nonce, tag) pairs.

4.2 ARC relations

We now refine the blind-issuance relations of Section 3 to the message split $\mu = \mathbf{m} \| \mathbf{k}$.

Issuance relation for ARC.. The public statement is

$$x_{\mathrm{IssueARC}} = (\mathsf{pp}_{\mathsf{Com}}, \mathbf{c}, r_0^I, r_1^I, T, B),$$

and the hidden witness is

$$w_{\mathrm{IssueARC}} = (\mathbf{m}, \mathbf{k}, r_0^H, r_1^H, \bar{r}, r_0, r_1, Z).$$

Define

$$\mathcal{R}_{\mathrm{IssueARC}} = \left\{(x_{\mathrm{IssueARC}}, w_{\mathrm{IssueARC}}) : \text{the following hold}\right\}$$

with constraints

$$\mathbf{c} = \mathbf{A}_{\mathsf{Com}}[(\mathbf{m}\|\mathbf{k})\|r_0^H\|r_1^H\|\bar{r}]^\top, \tag{13}$$

$$r_0 = \mathsf{center}(r_0^H + r_0^I), \tag{14}$$

$$r_1 = \mathsf{center}(r_1^H + r_1^I), \tag{15}$$

$$(B_3 - \mathbf{1}_n r_1) \odot Z = 1, \tag{16}$$

$$T = B_0 + B_1(\mathbf{m}\|\mathbf{k}) + B_2 r_0 + Z, \tag{17}$$

and membership/bound checks

$$\|\mathbf{m}\|_\infty,\ \|r_0^H\|_\infty,\ \|r_1^H\|_\infty,\ \|\bar{r}\|_\infty,\ \|r_0\|_\infty,\ \|r_1\|_\infty \le \beta_{\mathsf{msg}}, \qquad \mathbf{k} \in \mathcal{K}_{\mathsf{key}}.$$

In the bounded-key instantiation, the public parameters choose $\mathcal{K}_{\mathsf{key}} \subseteq \{\mathbf{k} : \|\mathbf{k}\|_\infty \le \beta_{\mathsf{msg}}\}$, so key membership can be implemented by the same bounded-membership gadget. Thus the same pre-sign proof simultaneously certifies commitment opening, centering, admissibility, and the BB-tran target for the full signed message.

Showing relation. The public showing statement is

$$x_{\mathrm{ShowARC}} = (\mathsf{nonce}, \mathsf{tag}, \mathbf{A}, B, \beta_{\mathsf{msg}}, \beta_{\mathtt{sig}}),$$

and the hidden witness is

$$w_{\mathrm{ShowARC}} = (\mathbf{m}, \mathbf{k}, r_0, r_1, Z, \mathbf{u}).$$

Define

$$\mathcal{R}_{\mathrm{ShowARC}} = \left\{(x_{\mathrm{ShowARC}}, w_{\mathrm{ShowARC}}) : \text{the following hold}\right\}$$

with constraints

$$(B_3 - \mathbf{1}_n r_1) \odot Z = 1, \tag{18}$$

$$\mathbf{A}\mathbf{u} = B_0 + B_1(\mathbf{m}\|\mathbf{k}) + B_2 r_0 + Z, \tag{19}$$

$$\mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce}), \tag{20}$$

$$\mathsf{nonce} \in \mathcal{N}, \tag{21}$$

and membership/bound checks

$$\|\mathbf{m}\|_\infty, \|r_0\|_\infty, \|r_1\|_\infty \le \beta_{\mathsf{msg}}, \qquad \mathbf{k} \in \mathcal{K}_{\mathsf{key}}, \qquad \|\mathbf{u}\|_\infty \le \beta_{\mathtt{sig}}.$$

In the bounded-key instantiation, $\mathbf{k} \in \mathcal{K}_{\mathsf{key}}$ is enforced by the corresponding bounded-membership constraints. This relation is the core of the ARC presentation proof: it certifies that the public tag was derived from the same hidden key that was signed during issuance.

4.3 ARC protocol

We define the anonymous rate-limiting credential scheme

$$\mathsf{ARC} = (\mathsf{Setup}, \mathsf{IssuerKeyGen}, \mathsf{Issue}, \mathsf{Show}, \mathsf{Verify})$$

using the blind-issuance protocol of Section 3 and the relations above.

$\mathsf{Setup}(1^\lambda)$ **and** $\mathsf{IssuerKeyGen}(\mathsf{pp}_{\mathsf{BSig}})$**:** These are the same as in Section 3.

Issue**:** The issuer $\mathcal{I}$ holds sk while the holder $\mathcal{H}$ has attribute vector $\mathbf{m}$ and verification key vk.

(1) The holder samples a secret PRF key $\mathbf{k} \xleftarrow{\$} \mathcal{K}_{\mathsf{key}}$ together with holder-side randomness $r_0^H, r_1^H, \bar{r}$, forms the full message

$$\mu = \mathbf{m}\|\mathbf{k},$$

and computes

$$\mathbf{c} = \mathbf{A}_{\mathsf{Com}}[\mu\|r_0^H\|r_1^H\|\bar{r}]^\top.$$

It sends $\mathbf{c}$ to the issuer.

(2) The issuer samples r_0^I, r_1^I coefficient-wise in $[-\beta_{\mathsf{msg}}, \beta_{\mathsf{msg}}]$ and returns them.

(3) The holder computes r_0, r_1 by centering. If $B_3 - \mathbf{1}_n r_1 \notin R_q^{\times n}$, it aborts the current issuance and restarts from the issuer-randomness round. Otherwise it derives $Z = (B_3 - \mathbf{1}_n r_1)^{-1}$ and sets

$$T = B_0 + B_1(\mathbf{m}\|\mathbf{k}) + B_2 r_0 + Z.$$

(4) The holder produces

$$\pi_{\mathsf{Issue}} \leftarrow \mathsf{NIZK.Prove}\big(w_{\mathrm{IssueARC}} \mid \mathcal{R}_{\mathrm{IssueARC}}, x_{\mathrm{IssueARC}}\big)$$

and sends $(T, \pi_{\mathsf{Issue}})$ to the issuer.

(5) The issuer verifies the proof. If it accepts, it samples

$$\mathbf{u} \leftarrow \mathsf{TDRH.SampPre}(\mathsf{td}, T)$$

and sends $\mathbf{u}$ to the holder; otherwise it aborts.

(6) The holder checks $\mathbf{Au} = T$ and $\|\mathbf{u}\|_\infty \le \beta_{\mathtt{sig}}$. If the checks succeed, it stores the credential witness

$$\mathsf{cred} = (\mathbf{u}, \mathbf{m}, \mathbf{k}, r_0, r_1).$$

$\mathsf{Show}(\mathsf{cred}, \mathsf{nonce})$**:** On input a stored credential

$$\mathsf{cred} = (\mathbf{u}, \mathbf{m}, \mathbf{k}, r_0, r_1)$$

and a public nonce nonce, the holder computes

$$Z = (B_3 - \mathbf{1}_n r_1)^{-1}, \qquad \mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce}),$$

then generates a showing proof

$$\pi_{\mathsf{Show}} \leftarrow \mathsf{NIZK.Prove}\big(w_{\mathrm{ShowARC}} \mid \mathcal{R}_{\mathrm{ShowARC}}, x_{\mathrm{ShowARC}}\big).$$

It outputs the presentation

$$\mathsf{pres} = (\mathsf{nonce}, \mathsf{tag}, \pi_{\mathsf{Show}}).$$

$\mathsf{Verify}(\mathsf{vk}, \mathsf{pres}, \mathsf{state})$**:** On input

$$\mathsf{pres} = (\mathsf{nonce}, \mathsf{tag}, \pi_{\mathsf{Show}}),$$

the verifier first checks π_{Show} against $\mathcal{R}_{\mathrm{ShowARC}}$. If the proof rejects, it outputs $(0, \mathsf{state})$. Otherwise it checks whether $(\mathsf{nonce}, \mathsf{tag})$ is already present in state. If so, it rejects; if not, it inserts $(\mathsf{nonce}, \mathsf{tag})$ into state and outputs $(1, \mathsf{state})$.

4.4 Security decomposition

The current draft does not prove Definition 2.27 in full. Instead, it records the policy-soundness and presentation-unlinkability consequences obtained from the present issuance layer, the showing proof, and the concrete PRF assumption.

Correctness. If issuance completes successfully, then the holder stores a witness satisfying

$$\mathbf{Au} = B_0 + B_1(\mathbf{m}\|\mathbf{k}) + B_2 r_0 + Z, \qquad (B_3 - \mathbf{1}_n r_1) \odot Z = 1.$$

For any nonce, the holder can therefore satisfy the showing relation, and completeness of the NIZK makes the verifier accept unless the local rate-limiting state rejects the pair $(\mathsf{nonce}, \mathsf{tag})$.

Proposition 4.1 (Conditional policy soundness)**.** *Assume the post-sign NIZK is knowledge sound, the interactive issuance layer is one-more unforgeable for the raw BB-tran credential witnesses extracted by* $\mathcal{R}_{\mathrm{ShowARC}}$*, and the PRF evaluation is deterministic and correct. Then after at most* Q *issuance interactions, any PPT adversary can cause more than* $Q|\mathcal{N}|$ *state-accepted presentations only with negligible probability.*

Proof. Knowledge soundness of the showing proof yields a witness $(\mathbf{m}, \mathbf{k}, r_0, r_1, Z, \mathbf{u})$ satisfying $\mathcal{R}_{\mathrm{ShowARC}}$. In particular,

$$\mathbf{Au} = h_{\mathrm{tran}, \mathbf{m}\|\mathbf{k}, (r_0, r_1)}(B) \qquad \text{and} \qquad \mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce}).$$

By the assumed one-more unforgeability of the issuance layer, at most Q distinct valid raw credential witnesses can be produced after Q issuance interactions, except with negligible probability. For each fixed credential witness and each fixed nonce, PRF correctness fixes a unique tag. Therefore a stateful verifier that rejects repeated $(\mathsf{nonce}, \mathsf{tag})$ pairs can accept at most one presentation per nonce per valid credential witness. This gives the bound $Q|\mathcal{N}|$; accidental tag collisions only reduce the number of accepted pairs. □

Proposition 4.2 (Conditional presentation unlinkability)**.** *Assume honest-issuer public-surface hiding of the issuance layer, zero knowledge of the showing proof, and Assumption 2.33 for keys sampled from* $\mathcal{K}_{\mathsf{key}}$*. Then, for distinct fresh nonces, presentations produced from the same credential are computationally indistinguishable from presentations produced from independently issued credentials, except for the intended linkage caused by nonce reuse.*

Proof. Honest-issuer public-surface hiding prevents the honest issuance transcript from revealing the committed message and holder randomness. Zero knowledge of the showing proof hides the witness $(\mathbf{u}, \mathbf{m}, \mathbf{k}, r_0, r_1, Z)$ during presentation. For distinct fresh nonces, Assumption 2.33 makes the values $\mathsf{F}(\mathbf{k}, \mathsf{nonce})$ computationally indistinguishable from random-function outputs at those inputs, hence tags do not reveal whether two presentations used the same hidden key. The only intended exception is nonce reuse, where determinism of the PRF causes the tag to repeat and rate limiting deliberately creates linkage. □

Attribute privacy and selective disclosure. In the core ARC relation of this paper, $\mathbf{m}$ remains hidden during showing. Predicate proofs or selective disclosure can be added by extending $\mathcal{R}_{\text{ShowARC}}$ with further constraints on $\mathbf{m}$, without changing the issuance protocol or the signed-message format.

5 SMALLWOOD PROGRAMMING MODEL

This section gives the SmallWood formulation of the ARC-SPRUCE NIZKs. The prover commits to packed witness rows, represented as low-degree row polynomials over a packing set Ω and masked on a pairwise-disjoint hiding set Ω'. The verifier then checks PACS-style constraints on those committed rows. We record here the semantic statement layers and the exact coefficient-domain consequences needed by the security arguments; concrete row inventories, gadget-level realizations, and proof-stack engineering are deferred to Appendix C.

5.1 Proof interface

Fix a packing set $\Omega = \{\omega_1, \ldots, \omega_s\} \subset \mathbb{F}_q$ and a disjoint hiding set $\Omega' = \{\omega'_1, \ldots, \omega'_\ell\} \subset \mathbb{F}_q \setminus \Omega$. A compiled witness is a matrix

$$W \in \mathbb{F}_q^{n \times s}.$$

For each row i, the prover interpolates a polynomial $P_i \in \mathbb{F}_q[X]$ of degree $< s + \ell$ such that $P_i(\omega_j) = W_{i,j}$ for all $\omega_j \in \Omega$, while the values on Ω' are random masks. The committed witness is therefore a family of row polynomials

$$P = (P_1, \ldots, P_n).$$

SmallWood proves families of constraints over these rows. Parallel constraints are slotwise identities

$$f_a\big(P_1(\omega), \ldots, P_n(\omega), \Theta_{a,1}(\omega), \ldots, \Theta_{a,t_a}(\omega)\big) = 0 \qquad \text{for every } \omega \in \Omega,$$

where $\Theta_{a,j}$ are public helper rows. Aggregated constraints are domain-summed identities

$$\sum_{\omega \in \Omega} g_b\big(P_1(\omega), \ldots, P_n(\omega), \Psi_{b,1}(\omega), \ldots, \Psi_{b,u_b}(\omega)\big) = 0.$$

In ARC-SPRUCE, the Ajtai commitment equation, centering equations, and BB-tran target identity are affine coefficient-wise constraints; the inverse-witness equation $(B_3 - \mathbf{1}_n r_1) \odot Z = 1$ is bilinear; and the PRF checks contain the nonlinear Poseidon2-style S-box constraints. Range, carry, packing-consistency, and norm gadgets may additionally use aggregated checks.

5.2 Issuance as a SmallWood statement

The public issuance statement is

$$x^{\text{SW}}_{\text{Issue}} = (\mathsf{pp}_{\mathsf{Com}}, \mathbf{c}, r_0^I, r_1^I, T, B),$$

and the hidden witness is

$$w^{\text{SW}}_{\text{Issue}} = (\mu, r_0^H, r_1^H, \bar{r}, r_0, r_1, Z).$$

The committed witness rows encode the message lanes, the two holder-side randomness blocks, the commitment-only randomness block, the long centered target-randomizer $r_0 \in R_q^{\ell_{r_0}}$, the inverse-side randomness r_1, the inverse witness Z, and the auxiliary carry rows used by the centering gadget. Semantically, the proof certifies the following claims.

Commitment opening. The public commitment opens to the same hidden message and holder-side randomness used later by the target:

$$\mathbf{c} = \mathbf{A}_{\mathsf{Com}}[\mu \| r_0^H \| r_1^H \| \bar{r}]^\top.$$

Thus the message and holder randomness are fixed before the issuer sends its challenge randomness.

Centering constraints. The final BB-tran randomness is obtained from the committed holder randomness and the public issuer randomness:

$$r_0 = \mathsf{center}(r_0^H + r_0^I), \qquad r_1 = \mathsf{center}(r_1^H + r_1^I).$$

In the compiled constraint system this is realized coefficient-wise through carry variables $\kappa_0, \kappa_1 \in \{-1, 0, 1\}$ satisfying

$$r_b^H + r_b^I - r_b = \kappa_b \cdot (2\beta_{\mathtt{msg}} + 1) \qquad \text{for } b \in \{0, 1\}.$$

These carry rows are range-constrained and packed together with the source rows.

Inverse-witness equation. The proof certifies that r_1 is admissible and that Z is its inverse witness:

$$(B_3 - \mathbf{1}_n r_1) \odot Z = 1.$$

This is the algebraic form used by the issuance proof; admissibility is thus enforced internally by the witness rather than merely checked externally.

Target identity. The public target is the BB-tran target corresponding to the same hidden message and centered randomness:

$$T = B_0 + B_1\mu + B_2 r_0 + Z.$$

Hence the pre-sign proof binds one coherent hidden witness simultaneously to the Ajtai commitment, the centered-randomness derivation, and the public target signed by the issuer.

5.3 Showing as a SmallWood statement

The public showing statement is

$$x^{\text{SW}}_{\text{Show}} = (\mathsf{nonce}, \mathsf{tag}, \mathbf{A}, B, \beta_{\mathtt{msg}}, \beta_{\mathtt{sig}}),$$

with verifier state handled outside the proof. The hidden witness is

$$w^{\text{SW}}_{\text{Show}} = (\mathbf{u}, \mathbf{m}, \mathbf{k}, r_0, r_1, Z).$$

The committed rows encode the short signature witness $\mathbf{u}$, the hidden message/key split $(\mathbf{m}, \mathbf{k})$, the BB-tran randomness (r_0, r_1) with $r_0 \in R_q^{\ell_{r_0}}$, the inverse witness Z, and the PRF companion rows.

Semantically, the proof certifies the following coefficient-domain statement:

$$(B_3 - \mathbf{1}_n r_1) \odot Z = 1, \tag{22}$$

$$\mathbf{A}\mathbf{u} = B_0 + B_1(\mathbf{m}\|\mathbf{k}) + B_2 r_0 + Z, \tag{23}$$

$$\mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce}), \tag{24}$$

together with the coefficient bounds on $\mathbf{m}, \mathbf{k}, r_0, r_1$ and the shortness bound on $\mathbf{u}$. Equation (23) is the direct witness-form BB-tran signature relation. Eliminating Z yields

$$(B_3 - \mathbf{1}_n r_1) \odot \big(\mathbf{A}\mathbf{u} - B_0 - B_1(\mathbf{m}\|\mathbf{k}) - B_2 r_0\big) = 1, \tag{25}$$

which is the coefficient-domain relation used later in the security arguments.

5.4 Replay / transform implementation

Some implementations check the showing relation in a transform or replay basis. Let

$$\mathcal{F}_{\text{rep}} : R_q \to \mathcal{T}$$

be the public replay map, with block projections $\mathcal{F}_{\text{rep},b}$. The prover then carries replay rows for the images of

$$\mathbf{Au}, \qquad \mathbf{m}\|\mathbf{k}, \qquad r_0, \qquad r_1.$$

The compiled relation enforces exact source-to-replay bridges

$$\widehat{T}_b = \mathcal{F}_{\text{rep},b}(\mathbf{Au}), \qquad \widehat{\mu}_b = \mathcal{F}_{\text{rep},b}(\mathbf{m}\|\mathbf{k}),$$
$$\widehat{r}_{0,b} = \mathcal{F}_{\text{rep},b}(r_0), \qquad \widehat{r}_{1,b} = \mathcal{F}_{\text{rep},b}(r_1),$$

and checks the replay residual

$$(B^{\Theta}_{3,b} - \widehat{r}_{1,b}) \odot \big(\widehat{T}_b - B^{\Theta}_{0,b} - B^{\Theta}_{1,b} \odot \widehat{\mu}_b - B^{\Theta}_{2,b} \odot \widehat{r}_{0,b}\big) = \mathcal{F}_{\text{rep},b}(1). \quad (26)$$

Here $B^{\Theta}_{i,b}$ denotes the public replay image of the relevant BB-tran constant or coefficient block.

Assumption 5.1 (Exact replay map). *The public replay map $\mathcal{F}_{\text{rep}}$ used by the showing implementation is additive, multiplicative on all products used in the compiled residual, and injective on the coefficient-domain values that appear in the showing relation. If a concrete replay representation is not injective, the compiled relation must include explicit linking constraints sufficient to recover the same coefficient-domain identities.*

Theorem 5.2 (Replay-space certification implies the coefficient–domain BB-tran identity). *Assume Assumption 5.1. Assume further that the compiled showing relation enforces the exact bridges above and the replay residual* (26). *Then the extracted coefficient-domain witness satisfies*

$$(B_3 - \mathbf{1}_n r_1) \odot \big(\mathbf{Au} - B_0 - B_1(\mathbf{m}\|\mathbf{k}) - B_2 r_0\big) = 1$$

in R_q.

Proof. Substitute the exact bridge equalities into the replay residual. Additivity and multiplicativity of $\mathcal{F}_{\text{rep}}$ transform the replay residual into the replay image of the coefficient-domain residual

$$(B_3 - \mathbf{1}_n r_1) \odot \big(\mathbf{Au} - B_0 - B_1(\mathbf{m}\|\mathbf{k}) - B_2 r_0\big) - 1.$$

Injectivity then forces this coefficient-domain residual itself to vanish, which is exactly the displayed identity. □

Corollary 5.3 (Abstract BB-tran witness consequence). *Under the assumptions of Theorem 5.2, the extracted witness additionally satisfies*

$$(B_3 - \mathbf{1}_n r_1) \odot Z = 1, \qquad \mathbf{Au} = h_{\text{tran},\mathbf{m}\|\mathbf{k},(r_0,r_1)}(B), \qquad \mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce}).$$

Proof. The inverse equation $(B_3 - \mathbf{1}_n r_1) \odot Z = 1$ is part of the compiled relation. By Theorem 5.2,

$$(B_3 - \mathbf{1}_n r_1) \odot \big(\mathbf{Au} - B_0 - B_1(\mathbf{m}\|\mathbf{k}) - B_2 r_0\big) = 1.$$

Combining this with the inverse equation identifies the bracketed term with Z, hence

$$\mathbf{Au} = B_0 + B_1(\mathbf{m}\|\mathbf{k}) + B_2 r_0 + Z = h_{\text{tran},\mathbf{m}\|\mathbf{k},(r_0,r_1)}(B).$$

The PRF equation is already part of the showing relation. □

5.5 PRF companion relation

The public tag is tied to the same hidden key carried in the signed message through a companion PRF relation. Selector constraints bind the PRF key lanes to the key rows $\mathbf{k}$ and enforce $\mathbf{k} \in \mathcal{K}_{\text{key}}$; checkpoint constraints certify the nonlinear Poseidon2-style S-box steps; linear layers and round constants are replayed from public parameters; and the final feed-forward and truncation constraints bind the public pair $(\mathsf{nonce}, \mathsf{tag})$.

A crucial invariant is that the PRF key rows are not separate from the signature/message rows: the compiled showing statement is a *single* committed witness family. If auxiliary alias rows are introduced for engineering convenience, they must be tied back to the original key rows by explicit copy constraints. This prevents a prover from using one hidden key for the BB-tran signature relation and another hidden key for the PRF relation.

5.6 Statement strength

The SmallWood proof hides a family of row polynomials, not merely a tuple of scalars. Random masks on Ω' statistically hide unopened row values, while the DECS/LVCS/PCS stack reveals only the evaluations needed to check the PACS constraints. Consequently, the verifier learns only the public statement: during issuance this is $(\mathbf{c}, r^I_0, r^I_1, T)$ together with public parameters, and during showing it is $(\mathsf{nonce}, \mathsf{tag})$ together with the public verification context.

The replay-space statement of Theorem 5.2 is therefore the only statement certified directly by the post-sign proof. The coefficient-domain BB-tran identity used in the security arguments is not assumed separately; it is a theorem-level consequence of the compiled proof relation and the exact bridge constraints.

Corollary 5.4 (Consequence of an accepting post-sign proof). *If the post-sign NIZK is knowledge sound for the compiled showing relation, then every accepting proof yields a witness*

$$(\mathbf{u}, \mathbf{m}, \mathbf{k}, r_0, r_1, Z)$$

satisfying the coefficient-domain BB-tran signature relation and the PRF-tag relation. In particular,

$$\mathbf{Au} = h_{\text{tran},\mathbf{m}\|\mathbf{k},(r_0,r_1)}(B) \qquad \text{and} \qquad \mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce}).$$

6 PARAMETERS AND INTEGRATION

This section explains how parameters interact across the ARC-SPRUCE stack: the Ajtai commitment, the BB-tran rational hash, the lattice preimage sampler, SmallWood, and the PRF. Detailed parameter tables and extended benchmark data are deferred to Appendix D.

6.1 Parameter dependencies

All components share the same base ring R_q and modulus q:

- the commitment matrix, signature map $\mathbf{A}$, and hash key B are defined over R_q;
- SmallWood constraints are evaluated over the base field $\mathbb{F}_q$ and therefore use the same modulus;
- the PRF is instantiated over $\mathbb{F}_q$ so that its constraints are native to the proof system.

The coefficient bound $\beta_{\mathtt{msg}}$ governs the message space, holder-side randomness, issuer-side randomness, centering map, and membership polynomials in the NIZKs.

6.2 Commitment and shared-randomness parameters

The Ajtai commitment binds the vector

$$[\mu \| r_0^H \| r_1^H \| \bar{r}]^\top.$$

Its hiding parameters must satisfy the leftover-hash condition of Lemma 2.6. The commitment-only randomness $\bar{r}$ and any additional committed randomness coordinates must provide enough min-entropy to statistically hide the full committed vector.

The centering map uses the interval $[-\beta_{\mathtt{msg}}, \beta_{\mathtt{msg}}]$ and modulus

$$\Delta = 2\beta_{\mathtt{msg}} + 1.$$

For correctness of the shared-randomness step, both holder and issuer contributions must lie in $[-\beta_{\mathtt{msg}}, \beta_{\mathtt{msg}}]$ coefficient-wise. Lemma 2.2 is then applicable coefficient-wise: for fixed holder randomness, the centered sum is uniform when the issuer contribution is uniform.

A second entropy budget is needed for target hiding. Write

$$r_0 \in R_q^{\ell_{r_0}}$$

for the long BB-tran target-randomizer. Let

$$X^N + 1 \equiv \prod_{j=1}^{f_{\mathrm{fac}}} \Phi_j(X) \pmod q, \qquad \deg \Phi_j = d_{\mathrm{fac}}, \qquad N = f_{\mathrm{fac}} d_{\mathrm{fac}}.$$

If the coefficients of r_0 are uniform over $[-\beta_{\mathtt{msg}}, \beta_{\mathtt{msg}}]$, the bounded-uniform cyclotomic leftover-hash bound gives

$$\varepsilon_{r_0} = \frac{1}{2}\sqrt{\left(\left(\frac{q^n}{(2\beta_{\mathtt{msg}}+1)^{\ell_{r_0}}}\right)^{d_{\mathrm{fac}}} + 1\right)^{f_{\mathrm{fac}}} - 1}.$$

The parameters must be chosen so that $\varepsilon_{r_0} \le 2^{-\lambda}$. Under this condition, $(B_2, B_2 r_0)$ is statistically close to (B_2, U) with $U \xleftarrow{\$} R_q^n$, and therefore

$$T = B_0 + B_1\mu + B_2 r_0 + (B_3 - \mathbf{1}_n r_1)^{-1}$$

statistically hides μ for every fixed admissible r_1; see Lemma 2.11 and Theorem 2.12. The inverse-side term is treated only as an additive offset, so the hiding proof uses no bounded-noise assumption on $(B_3 - \mathbf{1}_n r_1)^{-1}$.

6.3 Sampler parameters

The trapdoor sampler must output short signatures $\mathbf{u}$ with overwhelming probability and with a distribution close to a discrete Gaussian over the appropriate lattice coset. This requires choosing:

- ring degree N and modulus q,
- trapdoor quality,
- Gaussian widths for the sampler, and
- an acceptance predicate that yields the public verification bound $\beta_{\mathtt{sig}}$.

We use an NTRU-style trapdoor together with a Gaussian preimage sampler; details and tuning rules are given in Appendix B.

6.4 BB-tran rational-hash parameters

The BB-tran target is evaluated only when

$$B_3 - \mathbf{1}_n r_1 \in R_q^{\times n}.$$

Honest target derivation therefore restarts when this condition fails. The restart probability must be negligible or or explicitly accounted for as the rational-hash admissibility error δ. The two BB-tran randomness blocks play different roles: the long linear-side randomizer $r_0 \in R_q^{\ell_{r_0}}$ carries the entropy required by the cyclotomic-LHL hiding theorem, whereas the inverse-side randomizer r_1 is controlled only by admissibility and restart probability. No entropy lower bound on r_1 is used in the target-hiding proof.

The pre-sign proof certifies the witness form

$$(B_3 - \mathbf{1}_n r_1) \odot Z = 1, \qquad T = B_0 + B_1\mu + B_2 r_0 + Z.$$

The showing proof certifies the same relation with $T = \mathbf{A}\mathbf{u}$. Equivalently, it certifies

$$(B_3 - \mathbf{1}_n r_1) \odot \left(\mathbf{A}\mathbf{u} - B_0 - B_1\mu - B_2 r_0\right) = 1.$$

Membership checks for $[-\beta_{\mathtt{msg}}, \beta_{\mathtt{msg}}]$ are implemented via vanishing polynomials over $\mathbb{F}_q$, so one requires $\beta_{\mathtt{msg}} \ll q$ to avoid modular wrap-around artifacts.

Non-blind reduction parameters. The non-blind reduction imported from DKLW25 is parameterized by a norm bound β_{DKLW}, an inadmissibility bound δ, and the predicate/linear-independence side conditions of the BB-tran family. The public verifier bound $\beta_{\mathtt{sig}}$ should therefore dominate the relevant DKLW norm bound under the chosen norm comparison.

6.5 SmallWood parameters

SmallWood introduces parameters controlling packing, masking, and soundness:

- the packing size $s = |\Omega|$,
- the tail length ℓ for HVZK masking and degree enforcement,
- the spot-check count ℓ' and the number of masks ρ for PIOP soundness,
- the degree bounds for witness rows and constraint-derived masks, and
- the small-field knob θ when using an extension-field variant.

The pre-sign proof is dominated by the Ajtai opening, centering equations, inverse-witness check, and BB-tran target identity. The showing proof adds the signature shortness gadget, the BB-tran signature equation with $T = \mathbf{A}\mathbf{u}$, and the PRF checkpoint constraints. Increasing ℓ_{r_0} enlarges the witness width and the number of linear constraints, but it does not increase the nonlinear degree of the BB-tran relation itself.

6.6 PRF parameters

The PRF must be secure at the target security level, efficiently representable in constraints, and compatible with $\mathbb{F}_q$. We use a Poseidon2-style permutation with feed-forward and truncation. Its parameters include the state width t, round counts (R_F, R_P), MDS matrices, round constants, S-box exponent d, truncation length ℓ_{tag}, and key space $\mathcal{K}_{\mathrm{key}}$. We require $\ell_{\mathrm{tag}} \ge \lceil \lambda / \log_2(q) \rceil$ as an output-length condition; pseudorandomness of the resulting tags is exactly

Assumption 2.33 for keys sampled from $\mathcal{K}_{\text{key}}$. The output-length condition alone is not a proof of PRF security.

6.7 Sizing discussion

Concrete proof sizes depend jointly on the sampler parameters, the SmallWood masking parameters, the BB-tran relation, and the PRF checkpoint schedule. The issuance proof is smaller because the target T is public and the statement only checks consistency of the committed witness with that target. The showing proof additionally proves knowledge of a short preimage and a correct PRF evaluation.

A ADDITIONAL PRELIMINARIES

A.1 Security assumptions for Lattices

Fix a public left factor F, hint space H, target space G, predicate P, and query bound Q.

- **Hint-vSIS.** The adversary chooses a query set $\mathcal{Q} \subseteq H$ and a fresh target function $g^\star \in G \setminus \mathcal{Q}$ before seeing the sampled matrix A. It then receives short preimages of the queried hints $q(A)$ under $F(A)$ for every $q \in \mathcal{Q}$, and must output a non-zero short vector $u^\star$ such that $F(A)u^\star = g^\star(A)$. The instance is restricted by the predicate condition $P(F \cup \mathcal{Q} \cup (\{g^\star\} \setminus \{0\})) = 1$.
- **s-Hint-vSIS.** This is the strong variant of the same experiment. The adversary still chooses the hint set $\mathcal{Q}$ before seeing A, but it is allowed to choose the fresh target $g^\star$ only after seeing A and the hinted preimages.
- **s-\$-Hint-vSIS.** This is the strong random-hinted variant. It is the same as s-Hint-vSIS, except that the query set $\mathcal{Q}$ is sampled uniformly at random from H rather than chosen by the adversary.

  Now fix a keyed evaluation family $f : K \times M \times X \to R_q^n$.
- **GenISIS$_f$.** The challenger samples a public key $\kappa \leftarrow K$ and a uniform matrix $A \xleftarrow{\$} R_q^{n\times m}$. It provides a preimage oracle that, on input $\mu \in M$, samples $\chi \leftarrow X$, sets $t = f(\kappa, \mu, \chi)$, returns a short preimage s such that $As = t$, and records the tuple (s, μ, χ). The adversary wins if it outputs a fresh tuple $(s^\star, \mu^\star, \chi^\star)$ not previously returned by the oracle such that $As^\star = f(\kappa, \mu^\star, \chi^\star)$ and $0 < \|s^\star\| \le \beta$.
- **IntGenISIS$_f$.** This is an interactive version where the oracle includes the additional linear masking term and the stronger freshness condition used in DKLW's adaptive-security lifting.

The DKLW IntGenISIS$_f$-to-GenISIS$_f$ reduction is used only under the parameter conditions of their lifting theorem, including its ring, modulus, smoothing, dimension, and bound-loss requirements [4, Theorem 6].

A.2 Centering map and carries

Let BoundB $\in \mathbb{Z}_{>0}$ and $\Delta = 2\text{BoundB}+1$. Define the centering map center : $[-2\text{BoundB}, 2\text{BoundB}] \to [-\text{BoundB}, \text{BoundB}]$ by

$$\mathsf{center}(x) = \begin{cases} x & \text{if } x \in [-\text{BoundB}, \text{BoundB}], \\ x+\Delta & \text{if } x \in [-2\text{BoundB}, -\text{BoundB}-1], \\ x-\Delta & \text{if } x \in [\text{BoundB}+1, 2\text{BoundB}]. \end{cases}$$

Applied coefficient-wise, this map allows unbiased combination of two bounded values when at least one is uniform.

Lemma A.1 (Centering preserves uniformity)**.** *Fix any $a \in [-\text{BoundB}, \text{BoundB}]$. If $b \xleftarrow{\$} [-\text{BoundB}, \text{BoundB}]$ then* center$(a+b)$ *is uniform over* $[-\text{BoundB}, \text{BoundB}]$.

A.3 Vanishing polynomials for membership constraints

Membership constraints are expressed using vanishing polynomials over $\mathbb{F}_q$ with signed lifting.

Definition A.2 (Vanishing polynomial for $[-\text{BoundB}, \text{BoundB}]$)**.** Let BoundB $\in \mathbb{Z}_{>0}$. Define

$$P_{\text{BoundB}}(X) \;=\; \prod_{i=-\text{BoundB}}^{\text{BoundB}} \big(X - \langle i \rangle_q\big) \;\in\; \mathbb{F}_q[X].$$

For $x \in \mathbb{F}_q$, we have $x \in [-\text{BoundB}, \text{BoundB}]$ (via signed lifting) iff $P_{\text{BoundB}}(x) = 0$ in $\mathbb{F}_q$.

Definition A.3 (Vanishing polynomial for $\{-1, 0, 1\}$)**.** Define $P_1(X) = (X+1)X(X-1) \in \mathbb{F}_q[X]$. Then $P_1(x) = 0$ iff $x \in \{-1, 0, 1\}$.

B GAUSSIAN PREIMAGE SAMPLING

This appendix describes the Gaussian preimage-sampling interface used by the issuer to sign a target $t \in R_q^n$ by producing a short preimage $u \in R^m$ such that $Au = t$ in R_q^n. The security import in Section 2.5 requires the admissible trapdoor-suite interface of Definition 2.7. The concrete implementation candidate described here follows the vSIS–BB-tran setting [4] and uses an NTRU trapdoor generated via Antrag together with a hybrid two-plane sampler [5]. Antrag is used here as a trapdoor-generation and quality-tuning source, not as a standalone proof of the full DKLW admissible-suite interface.

B.1 The preimage sampling problem

Given a public linear map $A \in R_q^{n\times m}$ and a target $t \in R_q^n$, the signer samples

$$u \;\leftarrow\; \mathsf{SampPre}(\mathsf{td}_A, t) \qquad \text{such that} \qquad Au \equiv t \pmod q$$

and u is short (according to a public bound $\beta_{\mathtt{sig}}$). Equivalently, u lies in the coset lattice

$$\Lambda_\perp^t(A) := \{\, x \in R^m : Ax \equiv t \pmod q \,\}.$$

For compatibility with the DKLW import, the sampler layer must provide: (i) correctness ($Au = t$ and the norm test fails only with negligible probability), (ii) a preimage distribution close enough to the required lattice-coset Gaussian or an explicitly assumed substitute distributional guarantee, and (iii) a tail bound that justifies the public verification bound $\beta_{\mathtt{sig}}$ in the coefficient ℓ_∞-norm used by the ARC relations.

B.2 NTRU trapdoor form and quality

We work with the rank-2 NTRU module over the cyclotomic ring $R = \mathbb{Z}[X]/(X^N+1)$ and its quotient R_q. An NTRU public key has the form $h = gf^{-1} \bmod q$ for secret $f, g \in R$. The corresponding module is

$$\mathsf{L}_{\mathsf{NTRU}}(h) \;=\; \{(u, v) \in R^2 : uh - v \equiv 0 \;(\mathrm{mod} q)\}.$$

A trapdoor basis of $\mathsf{L}_{\mathsf{NTRU}}(h)$ is

$$B_{f,g} = \begin{pmatrix} f & F \\ g & G \end{pmatrix}, \qquad \text{with} \qquad fG - gF = q,$$

whose columns are short and satisfy $g/f \equiv G/F \equiv h \pmod q$. Following [5], we measure trapdoor quality via per-slot energies under the canonical embeddings φ_i, requiring a uniform window around q. Concretely, for a quality parameter $\alpha > 1$ we target

$$\frac{q}{\alpha^2} \le |\varphi_i(f)|^2 + |\varphi_i(g)|^2 \le \alpha^2 q \qquad \text{for all slots } i.$$

Smaller α corresponds to tighter trapdoors and, for fixed N, q, to shorter signatures for the same security.

B.3 Sampler definition and tuning

Our candidate sampler is a ring-based hybrid two-plane sampler in the evaluation domain. The sampler's main tuning parameters are:

- modulus q and ring degree N,
- trapdoor quality α,
- acceptance radius R_{acc} controlling rejection and norm tails,
- per-slot widths $\sigma_k[i]$ for the two planes $k \in \{1, 2\}$.

The concrete calibration formulas are implementation/tuning rules for the candidate sampler. The paper's signature-layer theorem uses them only through the admissible-suite properties in Definition 2.7: correctness, the required distributional hiding guarantee, and a tail bound implying $\|u\|_\infty \le \beta_{\mathtt{sig}}$ except with negligible probability.

Per-slot calibration. Let $\widetilde{b}_k[i]$ denote the per-slot Gram–Schmidt lengths for plane k at slot i. A standard calibration equalizes acceptance probability across slots:

$$\sigma_k[i] = \sqrt{\frac{\alpha^2 q\, R_{\mathrm{acc}}^2}{\|\widetilde{b}_k[i]\|^2} - R_{\mathrm{acc}}^2} \qquad (k \in \{1, 2\}).$$

This prevents weak slots and yields consistent output variance.

Hiding threshold (smoothing). To achieve statistical hiding, we enforce a floor $\sigma_{\min}$ such that $\sigma_k[i] \ge \sigma_{\min}$ for all k, i, where $\sigma_{\min}$ exceeds the smoothing parameter of the relevant lattice cosets. In practice, we apply a global scale factor to raise all $\sigma_k[i]$ above the theoretical floor with slack.

B.4 Annulus sampling (Antrag key generation)

Antrag replaces trial-and-error trapdoor generation with direct annulus sampling in the Fourier domain [5]. For each conjugate pair (there are $N/2$ independent slots), the key generator samples $(|\varphi_i(f)|, |\varphi_i(g)|)$ uniformly in an annulus arc $A^+(r, R)$ with

$$\sqrt{q}/\alpha < r < R < \alpha\sqrt{q},$$

chooses random phases, inverse-embeds to the coefficient domain, and rounds to obtain $(f, g) \in R^2$. By selecting a "middle" annulus window (see [5]) one ensures that the rounded trapdoor satisfies the desired α-window with high probability and with modest repetition.

B.5 Hybrid two-plane sampler (details)

We sketch the two-plane hybrid sampler in the notation of [5]. Let td_A encode an NTRU trapdoor for A and let $t \in R_q^n$ be the target. Write $\langle \cdot \rangle_q$ for signed coefficient lift and define center_q as coefficient-wise centering into $(-\frac{q}{2}, \frac{q}{2}]$.

Algorithm 1 $\mathsf{SampPre}(\mathsf{td}_A, t)$: two-plane hybrid sampler (sketch)

1: Lift t to coefficients and set $c \leftarrow \mathsf{center}_q(t) \in (-\frac{q}{2}, \frac{q}{2}]^n$.
2: Using td_A, compute per-slot widths $\sigma_k[i]$ (two planes) calibrated to trapdoor quality.
3: **repeat**
4: Sample slot-wise Gaussian pairs on the two planes, reconstruct a coefficient-domain candidate u.
5: Accept iff the norm predicate holds for u (e.g., ℓ_2 and/or ℓ_∞ bound).
6: **until** accepted
7: **return** u.

Public relations. Downstream verification and proofs use only the public relation $Au = t$ and shortness of u bounded by $\beta_{\mathtt{sig}}$. The value of $\beta_{\mathtt{sig}}$ must be derived from the sampler's accepted-output tail bound and converted to the coefficient ℓ_∞-norm convention used in the ARC relations.

C SMALLWOOD PROOF-STACK DETAILS

This appendix records schematic compiled surfaces used by the issuance and showing statements. It is descriptive: the semantic statements are those of Section 5, and final proof-size claims require the complete row inventory, constraint degrees, and soundness calculation for the instantiated relation.

C.1 Row inventories

Issuance rows. The issuance proof commits rows encoding:

$$\mathbf{m}, \mathbf{k}, r_0^H, r_1^H, \bar{r}, r_0, r_1, Z,$$

where the long target-randomizer satisfies $r_0 \in R_q^{\ell_{r_0}}$. The proof also carries carry rows for the centering equations and auxiliary rows for range membership. The active relations are

$$\begin{aligned}
\mathbf{c} &= \mathbf{A}_{\mathsf{Com}}[\mathbf{m}\|\mathbf{k}\|r_0^H\|r_1^H\|\bar{r}]^\top, \\
r_0 &= \mathsf{center}(r_0^H + r_0^I), \qquad r_1 = \mathsf{center}(r_1^H + r_1^I), \\
(B_3 - \mathbf{1}_n r_1) \odot Z &= 1, \\
T &= B_0 + B_1(\mathbf{m}\|\mathbf{k}) + B_2 r_0 + Z.
\end{aligned}$$

The key rows also enforce $\mathbf{k} \in \mathcal{K}_{\mathrm{key}}$. In the bounded-key instantiation this is implemented with the same signed-lift bounded-membership convention as the attribute rows.

Showing rows. The showing proof commits rows encoding:

$$\mathbf{u}, \mathbf{m}, \mathbf{k}, r_0, r_1, Z,$$

with the same dimension convention $r_0 \in R_q^{\ell_{r_0}}$, plus rows for the signature shortness gadget and the PRF companion relation. The active algebraic relations are

$$(B_3 - \mathbf{1}_n r_1) \odot Z = 1,$$
$$\mathbf{A}\mathbf{u} = B_0 + B_1(\mathbf{m}\|\mathbf{k}) + B_2 r_0 + Z,$$
$$\mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce}).$$

C.2 Commitment and centering constraints

The Ajtai opening constraint is linear over R_q and is compiled coefficient-wise into SmallWood rows. The centering map is represented by carry variables. With $\Delta = 2\beta_{\mathtt{msg}} + 1$, each coefficient satisfies

$$r_b = r_b^H + r_b^I - \Delta j_b, \qquad j_b \in \{-1, 0, 1\},$$

for $b \in \{0, 1\}$. Membership in the bounded interval and in $\{-1, 0, 1\}$ is enforced by the vanishing polynomials of Appendix A.

C.3 BB-tran constraints

The BB-tran constraints are expressed in witness form. Issuance checks the public target equation

$$T = B_0 + B_1(\mathbf{m}\|\mathbf{k}) + B_2 r_0 + Z$$

and the inverse equation

$$(B_3 - \mathbf{1}_n r_1) \odot Z = 1.$$

Showing uses the same inverse equation and replaces the public target by the signature surface $\mathbf{A}\mathbf{u}$:

$$\mathbf{A}\mathbf{u} = B_0 + B_1(\mathbf{m}\|\mathbf{k}) + B_2 r_0 + Z.$$

Equivalently, the showing proof implies the eliminated coefficient-domain relation

$$(B_3 - \mathbf{1}_n r_1) \odot \big(\mathbf{A}\mathbf{u} - B_0 - B_1(\mathbf{m}\|\mathbf{k}) - B_2 r_0\big) = 1.$$

C.4 PRF companion constraints

The PRF companion relation binds the public pair $(\mathsf{nonce}, \mathsf{tag})$ to the hidden signed key $\mathbf{k}$. The proof commits key-lane rows, selected S-box checkpoint rows, and any auxiliary rows needed to replay the Poseidon2-style linear layers between checkpoints. Selector constraints tie the key-lane rows to the signed key coordinates and enforce $\mathbf{k} \in \mathcal{K}_{\mathrm{key}}$; final feed-forward and truncation constraints tie the public tag to the resulting PRF output. Security of this keyed construction is the explicit PRF assumption of Assumption 2.33.

C.5 Single committed witness

All rows in one proof live under one SmallWood commitment. This ensures that the prover cannot use one hidden message for the Ajtai opening, a different message for the BB-tran target, and a third key for the PRF. In the showing proof, the same hidden key $\mathbf{k}$ is used simultaneously in the signature relation and the tag relation.

D EXTENDED PARAMETERS AND BENCHMARKS

This appendix provides extended parameter discussion and additional benchmark data.

D.1 Admissibility checklist

A parameter tuple is *admissible* for ARC-SPRUCE if it simultaneously satisfies:

(1) **Bound vs. modulus:** $\beta_{\mathtt{msg}} \ll q$ so membership checks on $[-\beta_{\mathtt{msg}}, \beta_{\mathtt{msg}}]$ do not wrap modulo q.
(2) **Commitment hiding:** the commitment-only randomness $\bar{r}$ makes the factorization-aware bound of Lemma 2.6 at most $2^{-\lambda}$.
(3) **Target hiding:** the long BB-tran randomizer $r_0 \in R_q^{\ell_{r_0}}$ makes the factorization-aware bound of Lemma 2.11 at most $2^{-\lambda}$.
(4) **Sampler interface:** the trapdoor/preimage layer satisfies Definition 2.7, including the public ℓ_∞ shortness bound $\beta_{\mathtt{sig}}$.
(5) **Rational-hash guard:** denominator invertibility failure is negligible or explicitly carried as the rational-hash admissibility error δ.
(6) **SmallWood degree/soundness:** the DECS/LVCS/PCS/PIOP degree bounds and soundness parameters are computed for the actual issuance and showing relations.
(7) **PRF key space and tag length:** the relation enforces $\mathbf{k} \in \mathcal{K}_{\mathrm{key}}$, the PRF assumption is made for this same key space, and ℓ_{tag} is large enough for the intended tag length.

D.2 Smoothing and sampler budgets

Let $\eta_\varepsilon(\Lambda)$ denote the smoothing parameter of a lattice Λ. A sufficient condition for statistical hiding in Gaussian preimage sampling is to choose Gaussian widths above a floor $\sigma_{\min} \geq \eta_\varepsilon(\Lambda_\perp(A))$, with slack (Appendix B). For the ambient lattice $\mathbb{Z}^{2N}$ one may use the classical bound

$$\eta_\varepsilon(\mathbb{Z}^{2N}) \;=\; \sqrt{\ln\!\big(4N(1+2^\lambda)\big)/\pi}.$$

In our tuning, trapdoor quality α and acceptance parameters are chosen so that the output shortness bound $\beta_{\mathtt{sig}}$ holds with overwhelming probability.

D.3 Example lattice parameters

Table 1 records one concrete instantiation compatible with the sizing discussion in Section 6.

Knob / Result	Symbol	Value (example)
Ring degree	N	1024
Modulus	q	1038337
Security target	λ	128
Trapdoor quality	α	≈ 1.23 (measured)
Gaussian width scale	σ	tuned per-slot (Appendix B)
Signature bound	$\beta_{\mathtt{sig}}$	derived from sampler tail bound

Table 1: Example lattice parameters (illustrative; see Appendix B for the sampling interface and tuning).

D.4 LHL design points for the long r_0-randomizer

For the measured parameter scale $N = 1024$, $q = 1038337$, and $\lambda = 128$, note that

$$q \equiv 1 \pmod{2N}.$$

Thus $X^N + 1$ splits completely over $\mathbb{F}_q$, so $d_{\text{fac}} = 1$ and $f_{\text{fac}} = N$. For a one-coordinate target ($n = 1$), Lemma 2.11 gives

$$\varepsilon_{r_0}(\beta_{r_0}, \ell_{r_0}) = \frac{1}{2}\sqrt{\left(1 + \frac{q}{(2\beta_{r_0}+1)^{\ell_{r_0}}}\right)^N - 1}.$$

The target-hiding condition is

$$\varepsilon_{r_0}(\beta_{r_0}, \ell_{r_0}) \le 2^{-\lambda}.$$

Equivalently, in the small-error approximation one needs roughly

$$\ell_{r_0} \log_2(2\beta_{r_0} + 1) \gtrsim \log_2 q + 2\lambda + \log_2 N - 2,$$

not $\log_2 q + 2\lambda/N$.

Table 2 records representative design points computed from the displayed factorization-aware bound.

β_{r_0}	minimum ℓ_{r_0} for $\lambda = 128$	membership degree
1	180	3
5	83	11
8	70	17
31	48	63

Table 2: Representative target-hiding design points for the fully split example $N = 1024, q = 1038337, n = 1$. These values use the factorization-aware cyclotomic LHL bound.

The table shows that the fully split example requires a much wider r_0 than a total-entropy-only calculation would suggest. Increasing β_{r_0} reduces the required width ℓ_{r_0}, but it increases the degree of the range-check polynomial.

D.5 SmallWood soundness terms

SmallWood's soundness is a combination of DECS polynomial binding, LVCS/PCS binding, PIOP sampling errors, and Fiat–Shamir/grinding terms [6]. For parameters $(N, s, \ell, \ell', \rho, \eta)$, field $\mathbb{F}$, and degree bounds $d_{\text{decs}} = s + \ell - 1$ and d_Q, schematic source-compatible terms include

$$\varepsilon_{\text{decs}} = \binom{N}{d_{\text{decs}}+2}|\mathbb{F}|^{-\eta} + \frac{Q_{\text{RO}}^2}{2^{2\lambda}}, \qquad \varepsilon_{\text{sum}} = |\mathbb{F}|^{-\rho}, \qquad \varepsilon_{\text{tail}} = \frac{\binom{d_Q}{\ell'}}{\binom{|S|}{\ell'}}.$$

The final knowledge-soundness bound for ARC-SPRUCE must instantiate the exact SmallWood-ARK theorem with the actual row inventory, the actual d_Q, the Fiat–Shamir query bounds, and the grinding parameters. Our issuance statement includes the bilinear inverse check $(B_3 - \mathbf{1}_n r_1) \odot Z = 1$ together with the linear target identity given public T, so d_Q must account for the inverse relation and the range/membership checks. The showing statement additionally depends on the PRF effective degree (Appendix E).

D.6 SmallWood knobs (example)

For the issuance and showing proofs discussed in Section 6, one example parameter tuple is:

$$(s, \ell, \ell', \rho, \theta, \eta) \;=\; (16, 25, 2, 2, 4, 19),$$

with masking degree d_Q derived from the instantiated constraint degrees. The pre-sign statement includes the bilinear inverse-witness constraint together with linear commitment, centering, and target equations, while the post-sign statement adds the PRF S-box composition on top of the same BB-tran inverse check.

D.7 Extended benchmark grid (SmallWood)

For additional context on how SmallWood knobs affect proof size and proving time, Table 3 reports an example sweep over packing and soundness parameters for standalone SmallWood-style statements. These figures are not a final ARC-SPRUCE proof-size claim unless the row inventory, constraint degrees, PRF checkpoint schedule, and Fiat–Shamir soundness terms are instantiated for the exact ARC relations.

ProofKB	Time (s)	Bits	NCols	ℓ	ℓ'	ρ	θ	η
9.32	0.266	133.44	6	22	2	2	4	17
10.07	0.228	146.93	4	24	2	2	4	17
11.52	0.250	133.22	4	28	2	2	4	17
14.06	0.630	198.01	4	34	8	5	2	26
14.17	0.654	192.26	6	34	8	5	2	26
14.24	0.670	198.48	4	34	8	6	2	26

Table 3: Example sweep over SmallWood knobs for standalone parameter exploration. Sizes exclude public parameters and are not a final ARC-SPRUCE proof-size claim.

E PRF DETAILS AND MISCELLANEOUS

This appendix records a concrete ZK-friendly PRF candidate compatible with the canonical ARC protocol. The construction is Poseidon2-style, but its use as a keyed PRF is an explicit assumption (Assumption 2.33).

E.1 PRF specification

We use a Poseidon2-style permutation with feed-forward and truncation. Let $k \in \mathcal{K}_{\text{key}} \subseteq \mathbb{F}_q^{\ell_{\text{key}}}$ be the secret key encoded by the signed $\mathbf{k}$, and let $n \in \mathbb{F}_q^{\ell_{\text{nonce}}}$ encode the public nonce. Let $t = \ell_{\text{key}} + \ell_{\text{nonce}}$ be the permutation width. Define

$$\mathsf{F}(k, n) \;=\; \mathrm{Tr}\big(P(k\|n) + (k\|n)\big),$$

where $\mathrm{Tr} : \mathbb{F}_q^t \to \mathbb{F}_q^{\ell_{\text{tag}}}$ outputs the first ℓ_{tag} lanes and P is a Poseidon2-style permutation [7].

E.2 Permutation structure

Let $d \ge 3$ be the smallest constant such that $\gcd(d, q-1) = 1$. Fix round counts R_F (external rounds, even) and R_P (internal rounds), round constants, and MDS matrices M_E, M_I. The permutation has the form

$$P \;=\; E_{R_F} \circ \cdots \circ E_{R_F/2+1} \circ I_{R_P} \circ \cdots \circ I_1 \circ E_{R_F/2} \circ \cdots \circ E_1.$$

Each external round applies the S-box to all lanes, while each internal round applies the S-box to lane 1 only. This structure reduces nonlinear constraints in the post-sign NIZK while retaining diffusion.

External and internal rounds. Let $x = (x_1, \ldots, x_t) \in \mathbb{F}_q^t$. For each external round $i \in [1, R_F]$, fix per-lane constants $(c_1^{(i)}, \ldots, c_t^{(i)}) \in \mathbb{F}_q^t$. For each internal round $i \in [1, R_P]$, fix a scalar constant $\hat{c}^{(i)} \in \mathbb{F}_q$. Then:

$$E_i(x) \;=\; M_E \cdot \big((x_1 + c_1^{(i)})^d, \ldots, (x_t + c_t^{(i)})^d\big)^\top,$$

$$I_i(x) \;=\; M_I \cdot \big((x_1 + \hat{c}^{(i)})^d,\; x_2, \ldots, x_t\big)^\top.$$

The matrices $M_E, M_I \in \mathbb{F}_q^{t \times t}$ are chosen so that they are invertible and provide sufficient diffusion (e.g., Cauchy-style MDS matrices as in Poseidon2).

E.3 Parameter generation

All PRF parameters are public. A typical parameter generator fixes:

- the base field modulus q (shared with the rest of the protocol),
- the S-box exponent d with $\gcd(d, q-1) = 1$,
- width $t = \ell_{\mathsf{key}} + \ell_{\mathsf{nonce}}$ and truncation ℓ_{tag},
- round counts R_F, R_P,
- MDS matrices M_E, M_I and round constants $\{c^{(i)}\}, \{\hat{c}^{(i)}\}$.

The truncation length is chosen to satisfy $\ell_{\mathsf{tag}} \geq \lceil \lambda / \log_2(q) \rceil$ as an output-length requirement. Pseudorandomness is not implied by this length condition; it is exactly the concrete assumption stated in Assumption 2.33 for $k \xleftarrow{\$} \mathcal{K}_{\mathsf{key}}$.

E.4 Arithmetization strategy (System A with checkpoints)

Status of the checkpoint schedule. Section 5.5 uses only the abstract interface of three PRF companion families: key binding, nonlinear checkpoint constraints, and final feed-forward/truncation constraints. The grouped schedule recorded in this appendix is a candidate realization for the present instantiation. Any final implementation must fix the exact round counts, checkpoint positions, and resulting effective degree used in the SmallWood soundness calculation.

To keep showing proofs compact, we commit only selected nonlinear checkpoints:

- **Key lanes.** We commit one row per key lane $\mathsf{Key}[i]$ and enforce equality to an encoding of signed $\mathbf{k}$ using selector constraints at fixed Ω-coordinates.
- **S-box outputs.** We commit only the S-box outputs at a checkpoint schedule (e.g., every internal round and periodically among external rounds). Linear wiring between checkpoints (add-round-constants, MDS multiplications) is recomputed by the verifier from opened evaluations and public constants.

This yields a System-A style constraint system where the dominant degree comes from composing at most a small number of unchecked S-box layers between checkpoints.

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